Implementation Regulations CER HZ

Bachelor

Information & Communication Technology

full-time

ISAT 30020

2026-2027

UNIVERSITY OF APPLIED SCIENCES

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Chapter 1 General provisions

1.1 General

1.1.1 Course and Examination Regulations

The HZ Course and Examination Regulations Bachelor programme (hereinafter: CER HZ) cover the core of education within the HZ. That document provides a general overview of all Bachelor degree programmes taught at the HZ. The CER HZ contains institution-specific provisions i.e., those that apply to the entire HZ. The programme-specific CER HZ Implementation Regulations (hereinafter: Implementation Regulations) is determined for each degree programme by the executive board each year.

1.1.2 Implementation Regulations

These Implementation Regulations apply to the CER HZ Bachelor programme full-time, part-time and dual.

1.1.3 Comparability of credits

The Dutch Higher Education and Research Act (WHW) as well as the CER HZ mention study credits. These Implementation Regulations, in addition to the term credits, also refer to ECTS (European Credits Transfer System), where 1 ECTS is equal to 1 credit and thus a study load of 28 hours (Article 7.4 paragraph 1 WHW).

1.2 Establishment and evaluation

1.2.1 Process

The process of establishment and evaluation of these Implementation Regulations is described in article 1.3 CER HZ.

1.2.2 Review

The programme committee evaluates the manner of implementation of the education and examination regulations and the Implementation Regulations in question every year (Article 1.3 CER HZ).

Chapter 2 Implementation regulations

2.1 Registration, prior educational entry requirements and admission policy

2.1.1 Overview of additional prior educational entry requirements

(Articles 2.2(A/B) and 2.3(A/B) CER HZ)

Students with a havo diploma				
Havo profiles:	**NT**	**NG**	**EM**	**CM**
Admissible:	Yes	Yes	Yes	Yes

Students with a vwo diploma				
Vwo profiles:	**NT**	**NG**	**EM**	**CM**
Admissible:	Yes	Yes	Yes	Yes

2.1.2 Deficiency investigation

(Article 2.4 CER HZ)

The programme sets no additional prior educational entry requirements. Students therefore cannot be deficient. As a consequence an investigation into deficiencies is not applicable.

2.1.3 Additional entry requirements

(Article 2.5 CER HZ)

The programme has no additional requirements for admission.

2.2 Structure of degree programme

2.2.1 Programme profile

(Article 3.2 CER HZ)

The ICT study programme profile is based on the Bachelor of ICT domain description of HBO-I (Applied Higher Educational ICT-programs) www.hbo-i.nl/domeinbeschrijving. The domain description is a functional qualifications framework for universities, focusing on the starting proficiency of future ICT professionals. It is a national framework for the final qualifications for graduates of Dutch programmes for higher professional education in the ICT domain at an Associate, Bachelor, and Professional Master degree level. Maintained by the HBO-I foundation, the domain description is periodically updated in collaboration with the business community and is established by The Netherlands Association of Universities of Applied Sciences.

As of this academic year the program will gradually convert to the new domaindescription version of 2023. Academic year 2026-2027 will contain the first year of the new curriculum which is mapped to the new domain description. In the following years, each year a new year of content will be implemented. The programme aims to be fully converted by the academic year of 2029-2030.

The domain description outlines relevant competencies, the breakdown of competencies into professional duties, and examples of characteristic professional situations of starting ICT professionals. These examples function as illustrations of elements of the model and create a clear connection with the professional practice. The domain description will be regularly modified and updated to keep up with the rapidly developing ICT field.

adaptable (adj.)

able to change or be changed in order to deal successfully with new situations.

The Adaptable IT Professional

The demand for competent IT professionals is growing across Europe. Within Zeeland, there is a clear need for IT professionals who can respond to challenges and opportunities shaped by local industry, as well as the region's social and economic context. While the needs of regional and national organisations may be specific, these are often shared more broadly at the international level34. Business and industry contexts that require IT differ widely, and the ever-evolving technological landscape and broad demand from the industry requires that the IT professional be *adaptable* within organisational contexts.

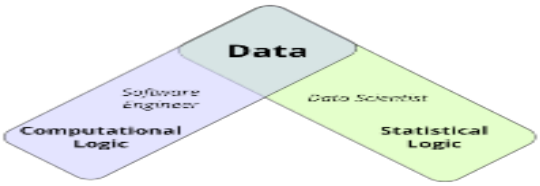
Vision on Professional Competence

The Adaptable IT Professional is rooted in foundational knowledge of information technology, which includes not only solving problems using basic programming concepts, but also key aspects of human-computer interaction and organisational structures in order to solve problems within a context. This knowledge will allow them to into small to medium enterprises, either an organisation with a small IT department, or a small to medium development house.

Upon entering the industry, they will exhibit proficiency within a set of skills, both technical and professional. Building on solid principles they can independently master the necessary software engineering and data science skills, technologies, and methodologies to evaluate problems, design relevant solutions, and critically evaluate implementations using a methodical, industry-standard approach.

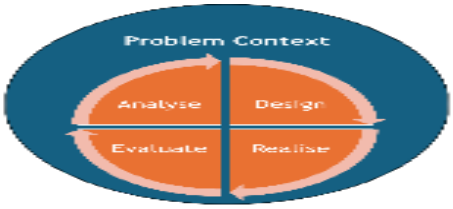
Following two internships at established organisations, Adaptable IT Professionals will immediately take to the ground running, understanding communication and collaboration with stakeholders, and immediately adding value to the organisation by realising actual solutions in the fields of software engineering or data science. By being comfortable with professional competences that allow them to understand and manage processes, the IT Professional will deliver relevant and authentic solutions within a wide range of problem and organisational contexts.

While the contexts may differ, problems stem from basic data transformation.



Data Transformation

With a focus on the computational logic of data, the professional can be seen as a software engineer focussing on algorithms and architecture. Focussing on statistical logic of the transformation of data, the professional can be seen as a data scientist. Skills are therefore required to transform to produce a useful platform or information using either computational and/or statistical logic. While the specific purpose of transforming data will differ, the understanding and basic transformation of data is still at the foundation. Different theoretical facets of computational and statistical knowledge will be instilled within the curriculum but also realised within a real-life project.



Project Cycle

IT professionals should be able to **analyse** the context to determine what the problem is that needs to be solved. From the analysis, a relevant solution should be **designed**. The designed solution should consider many different factors to achieve the most optimal design. The **realisation** of the design requires technical proficiency and should consider the existing environment. Finally, the solution should be **evaluated** to validate and verify the realised solution. This **process** should follow a well-structured methodology, ensuring each phase logically builds on the previous and provides a foundation for the next. This is the basis for all projects and is seen as the **Project Cycle**. The project cycle will also be the source for the learning outcomes.

Depending on the needs of the organisation, the job title will differ and independent of the job title and the context of the organisation, the same foundational skills are required.

Professional Profile

Regardless of the professional profile, the professional will need to deliver some product (comprised of multiple deliverables) that provides a solution within the specific problem context. The specific title will differ depending on the focus of data (be it algorithmic or statistical) that the professional will be working on.

Depending on the focus on the professional, the titles for graduates with an ICT degree typically include:

- Software Engineer: Designs, develops, tests, and maintains software systems, often focusing on problem-solving and quality.
- Back-end Engineer: Builds and maintains the server-side logic, databases, and APIs that power applications.
- Front-end Engineer: Develops and optimises the user interface and experience of websites or applications using web technologies.
- Data Engineer: Designs, constructs, and maintains data pipelines and infrastructure for efficient data storage and processing.
- Data Scientist: Uses statistical analysis, machine learning, and data modelling to extract insights and make predictions from data.

All these roles require that the professional have an understanding of architectures, business logic, data structures, and usability. Depending on the specific role in the organisation, the project of the problem context will shift. However, within the problem context the process that will be followed will be the same.

2.2.2 Learning outcomes

(Article 3.2 CER HZ)

The ICT program is offered in Dutch and English. These competences are according to the HBO-I domain description (see article 2.2.1) with data science as an addition. The profile matrix that is designed for the domain description contains three dimensions named in Table1.

Dimension	Represents

Activity • Analysis • Advise • Design • Realisation • Manage & Control	What does an ICT professional do?
ICT-architectural layers • User interaction • Organisational Processes • Software • Infrastructure • Hardware interfacing • Data science	Within which context?
Proficiency levels [1]	How complex is it?

Table 1: Dimensions of the domain description Bachelor of ICT

[1] The proficiency level is determined by the complexity of the context, the complexity of the content and level of autonomy involved in carrying out the assignment. A proficiency level is achieved when two of these facets reach the level concerned. For the third proficiency level, the autonomy and the complexity can be at level three of the context level, for example, while the complexity of the content is at level two. But it is also possible that the complexity of the context and the content are at level three while the autonomy is at level two. Further explanation of the four proficiency levels is in the domain description chapter 2.1.

Learning goals/outcomes

Students in Cohort prior to the current academic year (<2026), will continue with the curriculum applicable to them. In need of resits of year 1 they are eligible to resit the tests as mentioned in IR 2025-2026 for this academic year. Note: afterwards year 1 is fully phased out and students will then need to resit tests of the COHORT 2026 Curriculum.

Below is the total view of learning goals still applicable for COHORT <2026. All yellow lined learning goals are year 1 goals which are being phased out and have been transferred to the renewed curriculum, only resits of testing are based on these learning goals. The learning outcomes for Cohort 2026 and later can be found in Appendix 'Units of study'.

1. User interaction		
	1.1 analyse	
	B	You describe the important consequences for UX based on a target group analysis[B5] & [B6]
	1.2 Design	
	B	You describe UX test strategies suitable for a given situation[B5]
	1.3 Realise	
	A	You can apply design guidelines and corporate branding when realising a simple interaction within an information system [B3]
	B	You can realise a simple interaction within a team while taking into account consistency and standards [B3]
	C	You can help a user with preventing, recognising and solving erroneous actions in a consistent manner within a team [B4]
	D	You can help a user with recognising and solving erroneous actions [B4]
	E	You can apply standards and internal consistency when developing more complex functions within an application [B4]
	K	You describe the correct implementation of UX design choices [B5]
	L	You write a UX report accounting for design choices based on guidelines, human factors and/or emotional design.[B6]
	M	You test the UX of a product in a UX test report to evaluate the quality[B6]
	N	You recommend further development steps based on the UX test report[B6]
	1.4 Advise	
	A	You can draw up a datavision goal based on the project context and business goal taking into account the goal, the target group and the message. [B8]
	B	You can make a sound choice for a datavisualisation type suitable for the datavisualisation goal [B8]
	C	You can make a sound choice for visual elements suitable for the datavisualisation goal [B8]
	D	You can realise a datavisualisation based on sound research. [B8]

2. Organization processes		
	2.1 analyse	
	A	map, according to the given methodology, the current situation of a singular company process (IST) [B3]
	B	analyse the performance of an organization through a standard methodology. [B7]
	C	map an organization process of an existing organization by using suitable methodologies. [B7]
	D	you assess a given situation on various security aspects. [B7]
	E	you understand the importance of a sound BI report (B14)
	F	you understand what the necessity of BI is for companies (B14)
	G	You can independently make a validated process analysis for an ICT provision in the context of an internship [INTERNSHIP BIC]
	H	you clarify the company's current situation through coordinated KPIs and an obtained data set and you make an inventory of where the company can still take steps for improvement. Taking into account improvements in, among other things, new technologies.[S7]
	I	you map the branch and the company and you analyse how that process contributes to the company's goals [B7]
	J	You can independently make a validated process analysis (IST) for the ICT provisions in a complex context [S8]
	K	Students are capable of understanding the need for business to embrace data and can report what their maturity in this field is [B7]
	L	You analyze and evaluate the impact of Business Intelligence (BI) on the architecture of an organization from the perspective of Enterprise Architecture.
	2.2 Design	
	A	describe, according to a given methodology, a design for an improved company process through ICT (soll) [B3]
	B	you can map sound change strategies, so that you can choose the right strategy for the right change/company in a methodical way. (B13)
	C	you understand the ETL and the matching report process. (B14)
	D	You can independently make a validated process design and understand the relationship with the information provision in the context of an internship. [INTERNSHIP BIC]
	E	you analyse the IST of the processes within the company and you come up with realistic improvement proposals based on the various models and your own vision (SOLL). [S7]
	F	You can independently make a validated and considered process design (SOLL) in a complex context. [S8]
	2.3 Realise	
	A	you create KPIs for a dataset that you substantiate yourself and create a matching BI report. (B14)
	B	you carry out the entire process from importing the data to creating the report. (B14)
	C	you realise and evaluate an implementation (plan) based on your own design, so the company has a ready made plan to follow through with the implementation of the changes. [S7]
	D	you describe (and carry out if possible) a relevant change management method and strategy in which you help the employees with the changes they are about to encounter so that you can help resolve possible resistance. [S7]
	E	You independently realise an implementation(plan) and test the acceptance in a complex context. [S8]
	F	You show by a case that you apply the theory into practice [S7]

3. Infrastructure		
	3.1 analyse	
	3.2 design	
	A	The student can design a solution for a given project, making use of a cloud provider and taking into account the given preconditions.[B8]
	3.3 Realise	
	A	Make available a software system based on a Framework for users in a simple hosting environment [B4]
	3.4 Advise	
	A	The student can advise for a given project how it should be adapted to be able to use the functionalities of a cloud provider. [B8]
	3.5 Manage & Control	
	A	The student can select and employ and react accordingly on the generated metrics for a cloud application control tools. [B8]

4. Software		
	4.1 analyse	
	A	(group) you determine the systems context of a system to be developed [B3]
	B	(individually) you collect relevant data from one single requirement's source through a given elicitation technique [B3]
	C	(individually) you interpret collected data from the functional perspective to formulate and document requirements according to given standard method in natural language [B3]
	D	develop acceptation criteria for a user story [B4]
	E	you determine the System and the Systems context for a system to be developed with one interested party [B4]
	F	you collect information so as to formulate functional requirements for a system to be developed according to a standard method [B2]
	G	you document functional requirements for a system to be developed in natural language and in models through a given standard method [B4]
	M	you can map the trust boundaries of a complex system. (B14)
	N	You can independently make an analysis of a software engineering design problem in an internship context. [INTERNSHIP SE]
	O	you describe functional and quality specifications and limiting preconditions, in which at least maintenance and manageability are included in the local infrastructure and development processes. [S7]
	P	you use various types of sources and techniques for collecting specifications and preconditions. [S7]
	Q	You can validate the formulated specifications and preconditions and thus assess the degree of completeness and objectivity. [S7]
	R	You can thoroughly describe a technical and/or process-related problem concerning the production of software.
	S	You independently make an analysis of a software engineering design problem in a complex context [S8]
	T	You develop empathy for stakeholders to determine their challenges [B5]
	U	You create innovative ideas based on a defined problem [B5]
	V	You develop a prototype based on a validated idea [B5]
	W	You test the prototype extensively to come up with new insights. [B5]

	4.2 design	
	A	you design a database of a simple information system and document this by means a standard modelling technique [B3]
	B	You can make a functional design of a simple function of a system yet to be developed, and document it through a standard modelling technique. [B3]
	C	You can make a technical design of a simple function of a system as yet to be developed, and document it be means of a standard modelling technique [B3]
	D	you communicate more complex concepts and designs univocally with the professional field [B4]
	E	you write a technical description of (the internal) structure and working of an Object Oriented information system.[B2]
	F	You can solve a problem occurring in the market and involve the right stakeholders. [B6]
	G	you generate new insights by translating a solution into an MVP, test it, and analyse the metrics (results) [B6]
	I	you make a first overview of a business model. [B6]
	J	you describe the needs of the users of the software system to be developed.[B6]
	K	You master the basic concepts of (symmetrical) encryption, and argue for a given situation, how symmetrical encryption can improve the security of data.
	L	you determine the quality of the design, for example through testing or prototyping, taking into account the formulated quality characteristics (ISO 25010) [B6]
	M	you demonstrate the success of the solution in an organized way through metrics developed [B6]
	N	you write a techspecs report as reference that can be transferred to third parties [B6]
	O	you recognise and explain with which programming techniques you can solve certain software problems (B13)
	P	You can independently select, document, communicate and evaluate solutions for a software engineering design problem in an internship context using tests and prototypes [INTERSHIP SE]
	Q	you evaluate solutions based on the stated specifications and limitations (consistency) using tests, prototypes and comparable techniques. In addition, you analyse data collected with qualitative and/or quantitative analysis techniques. [S7]
	R	you select candidate solutions based on relevant, current and specialist professional knowledge from the ICT domain. [S7]
	S	you apply appropriate schematic techniques in the document where possible, which are in line with the chosen design strategy and goaled at the target group, which in any case consists of developers who (further) develop the product. [S7]
	T	You can independently select, evaluate (partial), document and communicate solutions for a software engineering design problem in a complex context. [S8]
	U	You can create measurable non-functional requirements for a given system [B5]
	V	You can design a substantiated architecture of a software system [B5]
	W	You can derive and interpret performance metrics for a given system [B5]

	4.3 Realise	
	A	You can realise a simple function within given concepts of a Framework [B3]
	B	You can test a software system based on a Framework on the own work environment [B3]
	C	Deliver Code that is acceptable for a production environment [B4]
	D	Within a given framework context apply a more complex concept [B4]
	E	Within a given organization and framework context develop an innovation [B4]
	F	you apply Object Oriented programming concepts to realise functionality.[B2]
	G	you apply programming concepts to realise functionality (Miller: 1. prescriptive, 2. applying) [B1] [B2]
	H	you write readable, well-organized code (Miller: 1. prescriptive, 2. applying) [B1] [B2]
	I	you make robust code (Miller: 1. prescriptive, 2. applying) [B1] [B2]
	J	Indicate for a given code example/class diagram which design patterns were applied. [B5]
	K	Apply a suitable design pattern for a given situation and work it out in both a class diagram and actual code. [B5]
	L	Recognise weak points in code, so-called code smells, and apply an appropriate standardised remedy, so-called refactoring. [B5]
	M	you apply the right combination of programming techniques for the problems in a complex software system.(B13)
	N	you perform a security audit through a given model. (B14)
	O	You can independently realise a suitable solution for a software engineering design problem in an internship context. [INTERNSHIP SE]
	P	you realise (prototypes of) a system existing of several sub systems and/or existing components [S7]
	Q	You can do research into the quality of the realised software such as functionality, security and performance. [S7]
	R	You independently realise a suitable solution to a software engineering design problem in a complex context, independently. [S8]
	S	You can implement a component for a given architecture [B5]

	4.4 Advise	
	A	You can independently give a suitable advice for solving a software engineering design problem in an internship context. [INTERNSHIP SE]
	B	you write a suitable advice on the results of a security research that was held. (B14)
	C	you explain the results of the security audit according to a model. (B14)
	D	you advise the customer on a solution for a software problem, convince the customer that the solution is in line with his/her objective and vision and you support the customer in the implementation of the solution or you give you process-oriented advice. [S7]
	E	You independently give a suitable advice for solving a software engineering design problem in a complex context. [S8]
	4.5 Manage & Control	
	A	You can organize and use tools to exchange code and documentation within a team [B3]
	B	Use the project tools to improve the process of analysis, design, realization, testing and making functions available in an application[B4]
	C	You can set up an environment on your working environment using virtualization and use it to test code. [B3]
	D	you set up (generic) servers to make an application available [B4]
	E	you use containerization to make an application available and modify it [B4]
	F	Master the advanced features of the distributed version control system (DVCS) Git to enable effective collaboration on a software project. [B5]
	G	Achieve manageability of your software project releases by choosing a branching model and corresponding workflow. [B5]
	H	Design a deployment pipeline that runs an existing open source software application and generates an automatic build. [B5]
	I	Proof your solution by performing a complete release from a change in code that generates corresponding executables executing all the steps of a release management cycle. [B5]
	J	Guarantee software quality by enabling quality tools and executing unit tests.[B5]
	K	you ensure confidentiality of a data set by applying cryptography [B7]

5. Hardware Interfacing		
	5.1 analyse	
	A	you describe the foundations of a computer system [B1]
6. Data Science (Cisp-DM Cycle)		
	6.1 You set up a data Science process	
	CRISP-DM phase(s): Business Understanding + Data Understanding	
	A	You can define and report the customers organisation and its problem [B7]
	B	You can define & provide data mining goals [B7]
	C	You can define business objectives and are aware of the need of information by the business [B7]
	D	You can collect provided data sets and make them usable for the data science process [B7]
	E	You describe collected and needed data by data types and metadata [B7]
	F	You define data mining goals success criteria [B8]
	G	You describe data mining activities based on choice of a basic machine learning model and relevant required activities [B8]
	H	You add extra self-organised and/or external data sources to the data science process [B8]
	I	You can compose a data management plan for a specific project, taking in account al facets of a given, recognised standard. (B14)
	J	You describe data mining activities based on choice of the best applicable machine learning model and relevant required activities [S7]
	K	You can independently set up a data science process in a internship context. [INTERNSHIP DS]
	L	You can independently set up a data science process in a complex context. [S8]

	6.2 You collect and address relevant data	
	CRISP-DM phase(s): Data Understanding + Data Preparation	
	A	You generate basic statistics summaries exploring data [B7]
	B	You create a basic quality description to validate relevant data [B7]
	C	You will exclude/include rows & columns to select relevant data [B7]
	D	You clean data in order to achieve correct data types and handle missing values [B7]
	E	You will perform basic feature extraction to construct correct and usable data [B7]
	F	You are capable of converting data in correct formats to visualize data [B7]
	G	You (re-)validate data after model generated assumptions [B8]
	H	You clean data by imputating and scaling relevant data [B8]
	I	You construct data by one-hot-encoding, defining targets & labelling relevant data [B8]
	J	You integrate relevant data by merging multiple data sources [B8]
	K	You convert data formats as prerequisite for relevant model(s) [B8]
	L	You validate data through statistical testing [S7]
	M	You imputate relevant values to the chosen data to substitute missing values [S7]
	N	You construct data by feature extracting (aggregates, target encoding) and/or unstructured data [S7]
	O	You integrate relevant data by merging & joining across multiple levels [S7]
	P	You convert data formats using sparse representation and include useful generators to enhance performance of your techniques [S7]
	Q	You independently collect and address relevant data in a internship context [INTERNSHIP DS]
	R	You independently collect and address relevant data in a complex context [S8]

	6.3 You perform data analysis	
	CRISP-DM phase(s): Modelling	
	A	You define metrics, independent records, & targets to generate a test design [B7]
	B	You build the model and benchmark the predictions with basic statistic tooling [B7]
	C	You assess relevant model(s) by the chosen metric [B7]
	D	You split data into test & train sets to generate a test design [B8]
	E	You build & train relevant model(s) and create predictions using the model(s) on test data set [B8]
	F	You assess the model(s) on chosen metrics of the defined success criteria [B8]
	G	You define a test design using cross validation & time splits [S7]
	H	You build a model taking feature selection, model tuning, bias, variance over/under fitting & learning curves into account [S7]
	I	You asses your model outcome using advanced metrics and graphical aids [S7]
	J	You can independently perform data analysis in a internship context. [INTERNSHIP DS]
	K	You can independently perform data analysis in a complex context. [S8]
	6.4 You evaluate & deploy results of the data science process	
	CRISP-DM phase(s): Evaluation + Deployment	
	A	You summarise and evaluate results with business objective(s) [B7]
	B	You set up a list of actions to determine following steps [B7]
	C	You produce a final report and present this to customer [B7]
	D	You review the data science process and you determine, and also report, lessons learned [B7]
	E	You evaluate and match success criteria with business objectives of the data science process [B8]
	F	You determine next steps and setup an advisory report for follow-up [B8]
	G	You produce a deliverable for customer [B8]
	H	You review the data science process and collect lessons learned on process & product [B8]
	I	You determine the next steps in a additional data science process cycle providing a conclusion supplemented with recommendations [S7]
	J	You advice the business successively implementing the data science process by a plan [S7]
	K	You can independently evaluate and deploy results of a data science process in a internship context. [INTERNSHIP DS]
	L	You can independently evaluate and deploy results of a data science process in a complex context. [S8]

7. Professional Skills		
	7.1 Professional Skills	
	M	you can employ the right professional skills to complete a project successfully in a complex environment [S7]
	N	you account for the choices made regarding the professional skills employed [S7]
	O	you can independently in a complex environment employ the right professional skills to complete a project successfully (S8)
	P	you account for the choices made regarding the professional skills employed. (S8)
	Q	You can function professionally in a company-related, ICT-related environment. [INTERNSHIP]
	7.2 show personal leadership. Year 1=Level 1 (Context: structured, predictable, known solution Content: Some of the basic concepts). Year 2 = Level 2 (Context: structured, unpredictable problem known solution space limited Contents: Several basic concepts and some in-depth concepts).	
	D	You are able to assess an ethical situation, following the steps of an ethical framework, and incorporate different schools of thought such as duty, virtue, and consequentialism ethics, in sound reasoning, i.e. avoiding logical fallacies.
	K	You can create a website as introduction to the program, include your motivation and show that you improve the website based on received feedback. Leading to a website that is improved in quality and attractiveness [B1]
	L	Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals, community goals and personal goals.
	M	Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals through participation in a project week.
	N	Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to personal goals through participation in an international week.
	O	You're considerate, see opportunities and seize them. You have a proactive attitude that you take initiative and feel responsible for what you do.
	P	You can motivate yourself and others, you are willing to help others / support (individual and team). You can present yourself or a team, take others into your own development.
	Q	You study demonstrates considered, strengthens your own learning and can recognize a learning need in yourself and mating act, reflect, evaluate, and give active feedback questions. You recognize when you need help and do it then.
	R	You can specify what type of professional you want to be and / or what type of positions you aspire, know your own strengths and weaknesses and can describe yourself well.

	7.3 Interact purposefully. Year 1=Level 1 (Context: structured, predictable, known solution Content: Some of the basic concepts). Year 2 = Level 2 (Context: structured, unpredictable problem known solution space limited Contents: Several basic concepts and some in-depth concepts).	
	A	you read IT-oriented English literature on HBO entrance and can extract the necessary knowledge from it
	B	you write IT-related English documentation on HBO entrance level, suitable for the message you want to convey and aimed at the target group.
	C	You focus on the various groups of stakeholders such as partners, interest groups, individual team members etc.
	D	You focus on what you want to communicate and what purpose you choose the most appropriate form and while you perform this proactively.
	E	You focus on your role in the context of the ICT job, you recognize these tasks and takes proactive. You dare others to speak (feedback) and is open to feedback. You are open to other opinions / views / arguments and see that as an enrichment. You consciously builds confidence in an interdisciplinary and intercultural cooperation context.
	F	~~you have mastered the Dutch (for Dutch track) or the English (for English and Dutch track) language in writing on level 3F(B2) (conditionally and thus tick off test within the course)~~
	H	You can read English for orientation (B2/C1)
	I	You can write formal English texts (B2/C1)
	J	You can give in English an verbal presentation
	K	you can communicate in a sound way with various departments within a company, taking into account hierarchical layers. (B13)
	L	As a project group you can report and present professionally, both verbally and in a report. [S7]
	M	As a project group you deliver structured products and account for everyone's role within the project, the method followed and evaluate the process and the product critically [S7]
	N	You can report and present professionally, both verbally and in a report [S8]
	O	You deliver structured products, account for the method followed and evaluate the process and the product critically. [S8]
	P	Students can present their project, the content of their portfolio and their process considerations in a sound way making plausible the equal contribution of each project member to the project.
	Q	as a team you can communicate your research in an organized way, appropriate for the audience.
	R	Students are able to deliver a solid product demonstration to the stakeholders in which they demonstrate the product and address the main challenges and present a realistic roadmap.

	7.4 Organize in a future-oriented way. Year 1=Level 1 (Context: structured, predictable, known solution Content: Some of the basic concepts). Year 2 = Level 2 (Context: structured, unpredictable problem known solution space limited Contents: Several basic concepts and some in-depth concepts).	
	L	Gives evidence that you are able to think ahead and plan ahead. You think methodically about the approach suitable for the assignment (identification of tasks, order of execution, proper prioritization) and how this contributes to the end result.
	M	You plan and monitors the time. You are cost conscious. You recognize opportunities and risks. You can thereby all time aware of agreements, legal regulations and ethical standards.
	N	You have a keen eye for the feasibility of duties in the organization. You taking into account the characteristics of the area of the assignment.
	O	You examine where necessary and relevant to the ethical implications of the tasks you perform. You recognize their own and others' limits and act accordingly.
	P	You can construct achievable and realistic goals within the time available which contribute to solving a problem or achieving a demand. The goals can be divided into multiple related detailed tasks.
	7.5 Solve problems in a research-oriented way. Year 1=Level 1 (Context: structured, predictable, known solution Content: Some of the basic concepts). Year 2 = Level 2 (Context: structured, unpredictable problem known solution space limited Contents: Several basic concepts and some in-depth concepts).	
	A	you can make a proposal for a sufficiently complex graduation assignment (B13)
	B	you can draw up a graduation plan for a complex graduation assignment. (B14)
	C	as a team you can deep dive in a new innovative technique/technology. Gaining knew knowledge by researching the way that is works and validate it by using an expert and reliable scientific resources.
	D	Gives evidence that your problems / challenges to identify and put in context (department / organization / business environment, social environment) and can analyse these problems. You are able, where appropriate and relevant to search for multiple solutions.
	E	Throughout the dissolution process you're curious, ask yourself if from different perspectives. You are pragmatically, creatively and critically and make if appropriate use of resources.
	F	You can make a thoughtful and methodical choosing the correct / most appropriate / suitable solution or approach. While you are critical about your own basis and used arguments.

2.2.3 Programme structure

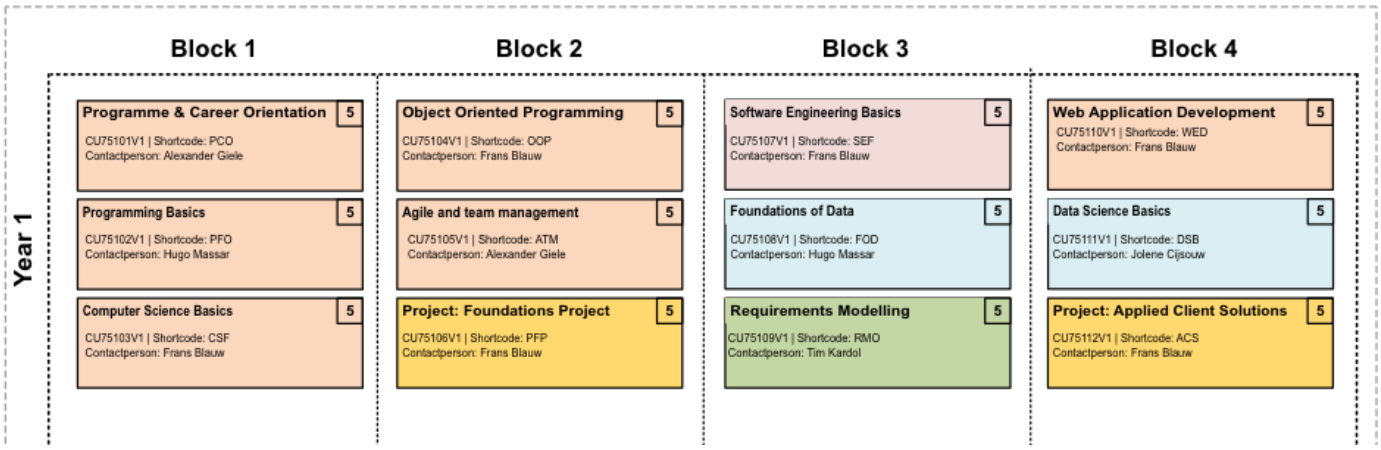
(Article 3.4 CER HZ)

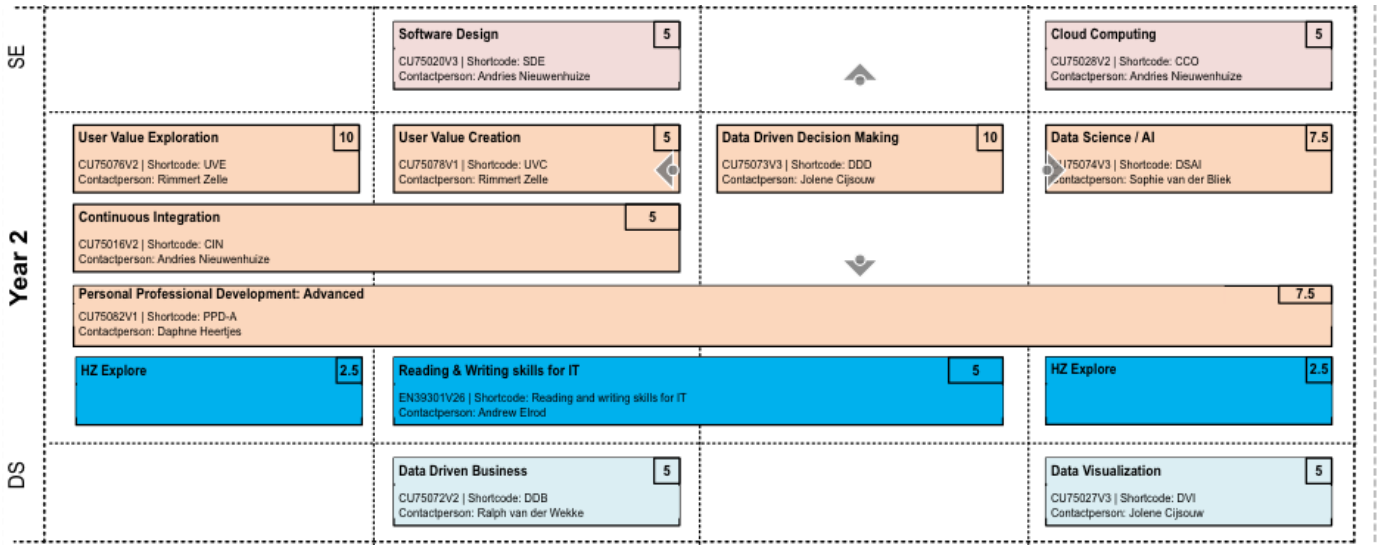
National name:	B ICT
International name:	B Information & Communication Technology
Orientation:	Bachelor
Title conferred:	Bachelor of Science
Programme duration:	240 study credits (ECTS)
Course workload propaedeutic phase:	60 study credits (ECTS)
Conclusion with propaedeutic examination:	Yes
Course workload main phase:	180 study credits (ECTS)
Variant:	full-time
ISAT code:	30020
Location:	Middelburg
Language:	Dutch and English
Effective date:	29-06-2018
Submission date:	01-11-2030
Joint degree programme:	Not applicable
180 credits (ECTS) fast track:	No

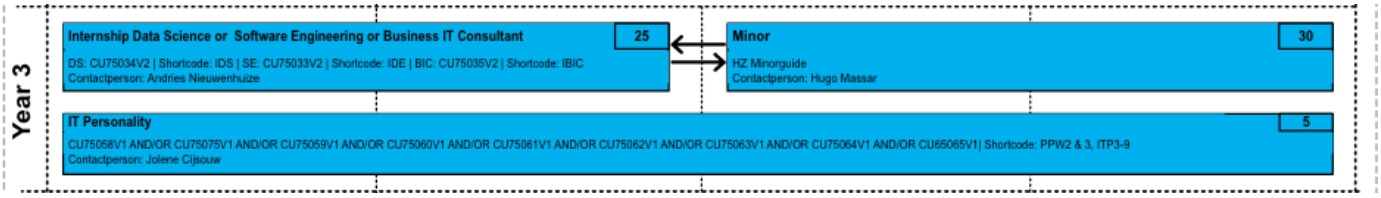
The degree programme is a coherent set of:	Full-time: Courses Part-time: N/A Dual: N/A
Possibility of learning path independence:	Full-time: N/A Part-time: N/A Dual: N/A

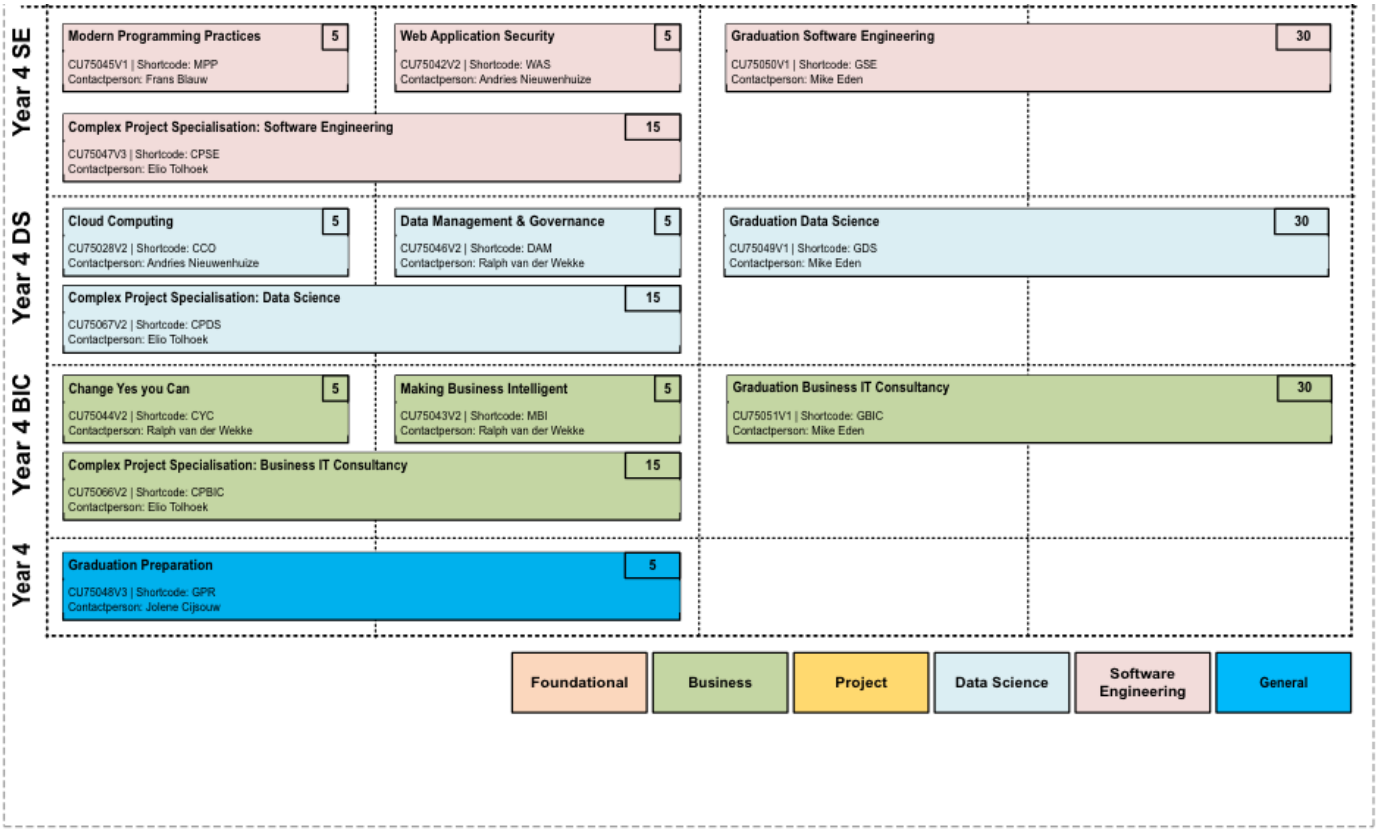
2.2.3a Examination programmes

(Article 6.1 CER HZ)









Note: The English for IT course (EN39301V24) is eligible for students on terms of an overall B2 level IELTS/Cambridge or completion of EN39002/EN39003/EN39004 based on the placement test results. The program has determined that the minimum level of English is B1. Therefore, based on the placement test, students either follow the Foundation Course (EN39002/EN39003/EN39004) courses or the EN39301V24 course. In any case, students are being prepared to address the critical need for improved reading comprehension and professional writing skills in the IT field. Additionally, students are free to follow both courses to enhance their English.

Furthermore, as a transition rule to include HZ Explore formally into the program, the HZ Explore courses (2x 2.5EC) in year 2 are elective with the IT Personality courses (4x 1.25EC) in year 3. Meaning that students must choose a total of 5 ECs from these courses. This transition rule applies to the cohort 2025/2026 only.

2.2.3b Transfer with an Associate degree certificate

(Article 3.4 CER HZ)

Students with a certificate from the following programme(s), awarded by Avans University of Applied Sciences (Brin 07GR), locations Roosendaal and 's-Hertogenbosch, are immediately admitted:

Ad Computer Science (Isat80075)

These students can during the first year of registration also register for the post propaedeutic phase of the programme. The institutional board exempts them from the requirement of having a certificate for the successful completion of the ‘propedeuse’ examination (via WHW Art. 7.30 paragraph 2). The Examination Board grants students with this certificate individual exemption from taking those examinations for which the Examination Board, prior to the first year of enrolment, has established, based on a programme comparison, that the student possesses the knowledge, insight and skills at the level that is being investigated through these examinations. The students must request exemption to this end in accordance with CER HZ Bachelor's degree programme full-time Article 4.6.

The above does not apply to students with the same certificate as that issued by an other university of applied sciences, nor does it apply to students with a certificate from an Associate degree programme other than the Associate degree programme(s) mentioned. The usual procedure for requesting exemptions applies to those students in accordance with the exemption policy.

2.2.4 Units of study propaedeutic phase

(Article 3.5.1 CER HZ)

See appendix 1.

2.2.5 Units of study main phase

(Article 3.5.1 CER HZ)

See appendix 2.

2.2.6 HZ Explore

(Article 3.5.5 CER HZ)

In the examination program of each bachelor's programme, 5 credits are reserved for HZ Explore electives. These credits are divided over the first 120 credits of the bachelor's programme.

HZ offers a range of activities from which students can choose to achieve the goals of HZ Explore. These activities have a study load of 1.25 or 2.5 EC. With this elective space, HZ offers students the opportunity to personalize and shape their own development during their studies by meeting and connecting. It also increases the possibilities for broadening across domains and stimulates broad social involvement. The range of HZ Explore activities is published annually in the HZ Explore Guide.

2.2.7 Specialisations

(Article 3.5.3 CER HZ)

The ICT programme offers 2 specific tracks. These are called study tracks. Each of these tracks consists of a compulsory part of a specific internship, a specific specialization semester and finally a specific graduation project. In addition, it is recommended to choose a matching minor. Specifically, it concerns the following tracks:

- Software engineering (SE)
- Data science (DS)

Cohort <2025

Students choose between two tracks in consultation with a lecturer/coach during year 2, block 1. The definite choice of one of these tracks will be during year 2, block 3. The study career coach is providing track specific information before the choice has to be made.

Afterwards, the choice will be definitive for the rest of the program. Obviously, students may choose to switch during their studies. The impactions though, need to be discussed with the study career coach.

Cohort >2026

Students choose between two tracks in consultation with a lecturer/coach during year 1, block 4. The study career coach is providing track specific information before the choice has to be made.

Afterwards, the choice will be definitive for the rest of the program. Obviously, students may choose to switch during their studies. The impact though, needs to be discussed with the study career coach.

Even though the tracks are labelled as specialisations, the degree is still for programme ICT and the chosen track is added to the degree ad an addendum, as well as the list of grades.

2.2.8 Internship

(Article 3.5.2 CER HZ)

The student is permitted to do their internship once the following requirements are met:

- the student is in possession of the propaeduetic certificate of the ICT programme
- the student has obtained at least 30 EC of completed courses in the second year of the program (Semesters 3 & 4)

Prior to the internship (which should take place either in semester 5 or 6) the student must have an approved Starting document, otherwise the startdate of the internship will not be acknowledged.

More information on our internship process is described in learn: https://learn.hz.nl/course/view.php?id=28894#section-0

Note: It is mandatory that students do their internship corresponding to their chosen track choice. Registering for a different internship will imply the student chooses for another track choice and thus needs to fulfil the applicable courses and finished track dependent courses will become extracurricular.

For information on the graduation/graduation internship, securing an internship and its assessment, please refer to the Graduation or internship course on learn which provides the student information and instruction.

2.2.9 Minor electives section

(Article 3.5.6 CER HZ)

The programme has a contiguous minor electives section with a total of 30 credits, which allows the student to give their own emphasis to their starting qualifications.

For the HZ-wide possibilities for filling in the minor electives section, the procedure for substantive approval and the minimum conditions for participation, see CER HZ Article 3.5.6.

If a student wishes to participate in a minor outside their own study program at a higher education institution or university in the Netherlands or abroad, prior permission from the partial examination board is required. The partial examination board checks whether the student has adequately justified the objectives and level of the minor to be chosen and whether the objectives and level of the minor to be chosen could not also be achieved by taking an HZ minor and whether the participation conditions are met as stated in article 3.6.5 CER HZ. Details are in the MyHZ info page.

2.2.10 Participation in international exchange programme

The conditions for participation are set out in Article 4.5(1) of the HZ Study Regulations. The ICT programme does not impose any additional conditions for participation.

Within the ICT programme there are opportunities to gain international experience during the internship, the minor or the graduation (blocks 9 & 10, 11 & 12, 15 & 16).

2.2.11 Graduation

(Article 3.5.2 CER HZ)

In order to participate in the graduation phase of the ICT programme (semester 8), the student has to have no more than 12,5 ECTS unpassed before the graduation (max 150 possible) of which the propaedeutic phase has been passed and granted.

The graduation phase is the last part of the programme and consists of 30 ECTS. The students have to do the graduation according to their chosen specialisation track:

- SE: CU75050V1 Graduation Software Engineering (GSE)
- DS: CU75049V1 Graduation Data Science (GDS)
- BIC: CU75051V1 Graduation Business IT Consultancy (GBIC)

These courses cannot be exempted.

For information on the graduation/graduation internship, securing an internship and its assessment, please refer to the Graduation or internship course on learn which provide the student information and instruction.

2.2.12 Assessment and inspection of results

HZ uses seven assessment types that are defined in the HZ Assessment Policy, namely:

- *Written knowledge test*; set of questions focused on knowledge reproduction and/or knowledge application, which are answered in writing.
- *Oral assessment*; set of questions about knowledge (application), which are answered orally.
- *Assignment*; representation of a performed (professional) task.
- *Presentation*; explanation or explanation before an audience of a performed (professional) task.
- *Portfolio*; collection of evidence of competence provided by the student.
- *Criterion-referenced interview*; discussion between assessor and student based on evidence provided in advance, using predefined criteria.

- *(Workplace) Assessment*; performance of (professional) tasks and/or skills (in an authentic context).

The Examination Board's fraud regulations and testing protocols apply to the taking of tests, see MyHZ.

The examiner ensures that the result of a test is registered in Osiris student within 15 working days after the student has taken the test and at least 5 working days before the next possibility for resit (Article 6.10 CER HZ).

The student has the right to inspect the assignments/questions, their elaborations and the assessment criteria of the test taken by the student within 10 working days after the date on which the result of the test was announced, or as much earlier as is necessary in connection with the next possibility of resitting the test (Article 6.5, 6.8 and 6.10 of the CER HZ).

2.2.13 Transitional arrangement

1. In general, the registered student is entitled to take tests for units of study that are no longer included in the curriculum, in the academic year following the academic year in which the unit of study was still included in the curriculum. The relevant tests (and resits) are taken during test and resit weeks of the unit of study in the academic year 2026-2027, unless otherwise agreed between the programme and students.
2. Compared to the academic year 2025-2026, changes have been made to test matrices for a number of units of study. This concerns the weighting, number and/or form of tests, test duration and relocation of tests to another unit of study. Without prejudice to the provisions of point 1, the responsible examiners of tests determine which tests students must take in 2026-2027 in order to meet the relevant requirements for tests from previous academic years. This means that in 2026-2027 the student cannot claim exactly the same test (form) from a previous academic year.
3. In cases where this transitional arrangement does not provide, the study career coach, programme manager and examination board will determine in consultation which effective alternative can be offered to enable completion of the exam programme.
4. The validity period of successfully completed examinations and tests may be limited if the tested knowledge, skills or insights are demonstrably outdated.

OLD			NEW			Remarks
Course	Title	Test	Course	Title	Test	
CU75001V3	Program- & Career orientation	Assignment	CU75101V1	Program- & Career orientation	Assignment	
CU75002V2	Computer Science Basics	Written test	CU75102V1	Computer Science Basics	Written test	
CU75003V1	Programming Basics	Assignment	CU75103V1	Programming Basics	Assignment	
CU75004V1	Object-Oriented Programming	Assignment	CU75104V1	Object-Oriented Programming	Assignment	
CU75004V1	Object-Oriented Programming	Group Project	CU75106V1	Project: Foundations Project	Portfolio	
CU75080V3	Framework Project 1	Assignment	CU75110V1	Web Application Development	Assignment	Assignments of Framework project 1&2 is converted into Web Application Development Assignment
CU75080V3	Framework Project 1	Project	CU75112V1	Project: Applied Client Solutions	Portfolio	
CU75011V4	Framework Project 2	Assignment	CU75110V1	Web Application Development	Assignment	
CU75011V4	Framework Project 2	Project	CU75112V1	Project: Applied Client Solutions	Portfolio	Projects of Framework project 1&2 is converted into Project: Applied Client Solutions
CU75068V3	Personal Professional Development Exploration	Portfolio	NA			Phases out: all contents of PPD-E are transitioned into: CU75101V1, CU75105V1, Cu75106V1, CU75107V1, CU75108V1, CU75109V1, CU75110V1, CU75111V1, CU75112V1. Resit is manually handled by exact copy of previous year test. No separate classes are provided.
CU75081V1	Business Understanding Basics		NA			Phases out: resit remains the same as previous year. No separate classes are provided.
CU75076V2	User Value Exploration	Portfolio	CU75076V3	User Value Exploration	Interview	The portfolio test has been replaced by a Criterion Referenced Interview. Weight changed from 40% to 50%
CU75076V2	User Value Exploration	Assessment	CU75076V3	User Value Exploration	Presentation	The assessment has been replace by a presentation. Weight changed from 60% to 50%

2.3 Education

2.3.1 Planning the learning pathway

The student follows an individual learning pathway. Before each semester or block, the student plans which units of study and tests will be part of the learning pathway. The student is obliged to the organisation of the degree programme, to the sequence and planning of the units of study and tests of the examination programme (elaborated in Article 2.2 of these implementation regulations) to which the student is linked and to the conditions for participation and registration (see Article 2.3.2 and Article 2.3.3 of these implementation regulations).

The student is guided in planning the learning pathway and making study choices.

2.3.2 Participation in education

The student can participate in a unit of study if the student meets the conditions for participation and is registered in Osiris for that unit of study.

The conditions for admission to and participation in specific examination programmes and parts of the degree programme determine which study units may be participated in. The relevant conditions for admission to and participation in specific examination programmes and parts of the course are described in Articles 2.2.3 and 2.2.6 to 2.2.11 of these implementation regulations.

Additional conditions for participation may be imposed per unit of study. These conditions for participation are indicated for the relevant units of study in Articles 2.2.4 and 2.2.5 and/or the tables in the appendix(es) of these implementation regulations.

2.3.3 Registration for units of study and tests

See CER HZ Article 4.4.

2.3.4 Osiris Student

More information about Osiris Student can be found on HZ Learn under Student - OSIRIS Support.

2.4 Study recommendation

The rules and procedure for issuing study recommendations are described in Chapter 8 of the CER HZ.

2.5 Experimental article

(Article 9.4 CER HZ)

The program does not conduct experiments in study year 2026-2027.

Chapter 3 Establishment

3.1.1 Term

The term of these implementation regulations is the same as the term of the CER HZ 2026-2027.

3.1.2 Date of approval

The Programme Committee has approved this implementation regulation on 02-07-2026.

3.1.3 Date of establishment

These implementation regulations were established by the Executive Board on 07-07-2026.

Appendix 1 Units of study propaedeutic phase

Block / Semester: B1					
CU75101V1	**Title: Programme- & Career Orientation**				
Course information					
Amount of study credits: 5			**Language:** English Dutch		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will discover your motivation for IT and set your goals for learning and growing as an IT professional.					
Course learning outcomes: You reflect on your motivation, strengths, and interests in relation to the professional field of IT through the realisation of a personal development resource within the context of higher education.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assignment	100%	5.5	B1.8, B2.4

Block / Semester: B1					
CU75102V1	**Title: Programming Basics**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will learn the basics of programming using JavaScript. You will learn the structure of a software application and the different ways to control the flow.					
Course learning outcomes: You apply fundamental programming concepts such as variables and control structures to realise simple software programs that demonstrate industry standards for good quality.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assignment	100%	5.5	B1.8, B2.8

Block / Semester: B1					
CU75103V1	**Title: Computer Science Basics**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will learn the basics of what makes a computer function. This knowledge will be the foundation for all computer-related learning.					
Course learning outcomes: You define and describe fundamental concepts of computer science.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Written knowledge test	Written Knowledge Test	100%	5.5	B1.8, B2.9

Block / Semester: B1 **Block / Semester: B2**					
CU75104V1	**Title: Object Oriented Programming**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will continue learning the basics of programming, but using TypeScript as your introduction into Object Oriented Programming. You will use these principles to develop a basic game engine and develop small games.					
Course learning outcomes: You apply fundamental object-oriented programming principles, such as abstraction, inheritance, and polymorphism to realise simple software solutions that demonstrate industry standards for good quality.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assignment	100%	5.5	B2.8, B2.10

Block / Semester: B2					
CU75105V1	**Title: Agile and Team Management**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will learn how an IT project is managed and operates indiviudally as well as within a team.					
Course learning outcomes: You define and describe fundamental project management processes, concepts, and methodologies, including teamwork principles, as applied to industry-standard IT projects.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assignment	100%	5.5	B2.8 & B2.10

Block / Semester: B2					
CU75106V1	**Title: Project: Foundations**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will complete your first project for a given target audience, primary school children. You will develop an educational game using stakeholholder and what you learnt so far.					
Course learning outcomes: You apply an iterative development methodology, collaborating in a team, to design, realise, and evaluate a simple software solution that meets a prescribed specification and demonstrates industry standards for good quality.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	5.5	B2.9 & B3.1 to B3.10

Block / Semester: B3					
CU75107V1	**Title: Software Engineering Basics**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will learn the foundations of sofware engineering. You will use architecture and design patterns to develop robust, sustainable software.					
Course learning outcomes: You define and apply fundamental software engineering principles, architectures, design patterns, and methodologies used to anlyse, design, and realise software systems using well-documented practices.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assignment	100%	5.5	B3.8, B3.10

Block / Semester: B3					
CU75108V1	**Title: Foundations of Data**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will be introduced to the data life cycle. You will learn to design relational databases using SQL and apply CRISP-DM principles to assess and clean data using Python, bridging the gap between software development and data science.					
Course learning outcomes: You define and apply foundational data modeling and preparation techniques to design relational structures and resolve data quality issues, to ensure data is fit for purpose in both software engineering and basic data analysis.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assignment	100%	5.5	B3.8, B3.10

Block / Semester: B3					
CU75109V1	**Title: Requirements Modelling**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will learn to elicitation and requirements modelling techniques to describe what is needed from a software project.					
Course learning outcomes: You define and use fundamental requirements modelling techniques used to analyse and document system requirements clearly and effectively using industry best practices.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assignment	100%	5.5	B3.8, B3.10

Block / Semester: B4					
CU75110V1	**Title: Web Application Development**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will use PHP to develop a web application framework from scratch. You will use your knowledge from software engineering as well as data science to build a industry-standard framework.					
Course learning outcomes: You integrate software engineering and data science principles and techniques to design and realise a robust, maintainable, and user-oriented web application, given a prescribed specification.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assignment	100%	5.5	B4.8, B4.10

Block / Semester: B4					
CU75111V1	**Title: Data Science Basics**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course, you will be introduced to the workflow of a Data Scientist using the CRISP-DM framework. Building on your data preparation skills, you will perform exploratory data analysis (EDA)to find patterns and use Python to build and evaluate baseline and basic regression and classification models.					
Course learning outcomes: You define and apply fundamental data science principles and methodologies to understand, transform, and visualise data, to build and evaluate simple predictive models using appropriate techniques.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assignment	100%	5.5	B4.8, B4.10

Block / Semester: B4					
CU75112V1	**Title: Project: Applied Client Solutions**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: Passed CU75103V1 (Object Oriented Programming) & CU75106V1 (Project: Foundations)					
Conditions for test participation: none					
Brief description of course content: In this course, you will work in a team to build a solution for a given client. You will analyse their problem, design and realise an appropriate solution. You will use an agile project management processes to manage your project.					
Course learning outcomes: You collaborate in a team using an iterative development methodology to apply basic requirements modelling, software engineering, and data science techniques to analyse, design, realise, and evaluate a simple software solution that demonstrates industry standards for good quality.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	5.5	B4.9 & B4.10

Block / Semester: S2					
CU75081V1	**Title: Business Understanding Basics**				
Course information					
Amount of study credits: 2.5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: The aim of these learning objectives and content is to introduce first-year IT students to the role of a business IT consultant and provide them with fundamental skills and concepts necessary for success in this role. By offering both theoretical knowledge and practical application, students can gain a better understanding of the work of a business IT consultant and develop their own skills in this field.					
Course learning outcomes: 2.1M: Acquire practical skills for conducting an IT consultancy project					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Video	100%	5.5	S2.19 & S2.20

Block / Semester: S1 **Block / Semester: S2**	
CU75068V3	**Title: Personal Professional Development Exploration**
Course information	
Amount of study credits: 12.5	**Language:** English Dutch
Conditions for course participation: none	
Conditions for test participation: 2 level checks submitted on time for feedback. If you have not submitted this, the first summative opportunity will be used for a formative submission and the Summative grading will follow upon resit.	
Brief description of course content: This course focuses on the four bachelor competences, interact purposefully, future oriented organising, research oriented problem solving and personal leadership. The course is alligned with the OOP project and framework project to practice the obtained skills.	
Course learning outcomes: 7.2O: You're considerate, see opportunities and seize them. You have a proactive attitude that you take initiative and feel responsible for what you do 7.2P: You can motivate yourself and others, you are willing to help others / support (individual and team). You can present yourself or a team, take others into your own development. 7.2Q:You study demonstrates considered, strengthens your own learning and can recognize a learning need in yourself and mating act, reflect, evaluate, and give active feedback questions. You recognize when you need help and do it then. 7.2R: You can specify what type of professional you want to be and / or what type of positions you aspire, know your own strengths and weaknesses and can describe yourself well 7.3C: You focus on the various groups of stakeholders such as partners, interest groups, individual team members etc. 7.3D: You focus on what you want to communicate and what purpose you choose the most appropriate form and while you perform this proactively 7.3E: You focus on your role in the context of the ICT job, you recognize these tasks and takes proactive. You dare others to speak (feedback) and is open to feedback. You are open to other opinions / views / arguments and see that as an enrichment. You consciously builds confidence in an interdisciplinary and intercultural cooperation contex 7.4L: Gives evidence that you are able to think ahead and plan ahead. You think methodically about the approach suitable for the assignment (identification of tasks, order of execution, proper prioritization) and how this contributes to the end result. 7.4M:You plan and monitors the time. You are cost conscious. You recognize opportunities and risks. You can thereby all time aware of agreements, legal regulations and ethical standards 7.4N: You have a keen eye for the feasibility of duties in the organization. You taking into account the characteristics of the area of the assignment. 7.4O: You examine where necessary and relevant to the ethical implications of the tasks you perform. You recognize their own and others' limits and act accordingly 7.4P: You can construct achievable and realistic goals within the time available which contribute to solving a problem or achieving a demand. The goals can be divided into multiple related detailed tasks 7.5D: Gives evidence that your problems / challenges to identify and put in context (department / organization / business environment, social environment) and can analyse these problems. You are able, where appropriate and relevant to search for multiple solutions. 7.5E:Throughout the dissolution process you're curious, ask yourself if from different perspectives. You are pragmatically, creatively and critically and make if appropriate use of resources 7.5F:You can make a thoughtful and methodical choosing the correct / most appropriate / suitable solution or approach. While you are critical about your own basis and used arguments.	
Assessment information	

Test code	Assessment type	Assessment description	Weighting Factor (%)	Minimum score	Test opportunities (block codes)
TOETS01 (VT)	Criterion-referenced interview	Criterium focused interview	100%	5.5	S2.11 to S2.20[1]

[1] Both opportunities are planned within the period of S2.11 to S2.20 in agreement with the student

Appendix 2 Units of study main phase

Blok / Semester: B1					
CU75076V3	**Titel: User Value Exploration**				
Cursusinformatie					
Aantal studiepunten: 10			**Voertaal:** Engels Nederlands		
Voorwaarden voor deelname cursus: none					
Voorwaarden voor deelname aan een of meerdere toetsen: none					
Beknopte beschrijving van cursusinhoud: User centred focus on exploring a problem context, setting up an architecture and understanding the user.					
Leerresultaten van de cursus: **Test 1:** You describe the important consequences for UX based on a target group analysis You describe UX test strategies suitable for a given situation You describe the correct implementation of UX design choices you describe the needs of the users of the software system to be developed you write a techspecs report as a reference that can be transferred to third parties You can create measurable non-functional requirements for a given system You can design a substantiated architecture of a software system You can derive and interpret performance metrics for a given system You can implement a component for a given architecture **Test 2:** You develop empathy for stakeholders to determine their challenges You create innovative ideas based on a defined problem You develop a prototype based on a validated idea 4.1W: You test the prototype extensively to come up with new insights					
Toetsgegevens					
Toetscode	**Toetsvorm**	**Omschrijving**	**Weging** (%)	**Bodem cijfer**	**Toetsgelegenheden** (blokcodes)
TOETS01 (VT)	Criteriumgericht interview	Interview	50%	5,5	S1.8 & S1.10
TOETS02 (VT)	Presentatie (individueel)	Presentation	50%	5,5	S1.2 & S1.10

Block / Semester: S1					
CU75016V2	**Title: Continuous Integration**				
Course information					
Amount of study credits: 5			**Language:** English Dutch		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: Develop a thorough understanding of a version control system (VCS) and learn strategies to incorporate a VCS in effective team collaboration. Setup a complete CI pipeline with an automated build for a given project. Add tests and metric tools like code coverage to control the software quality. Course will be based on several deliverables. The course planning is based on different types of releases. Improved by feedback each deliverable will be part of the final portfolio.					
Course learning outcomes: 4.5F: Master the advanced features of the distributed version control system (DVCS) Git to enable effective collaboration on a software project 4.5G: Achieve manageability of your software project releases by choosing a branching model and corresponding workflow 4.5H: Design a deployment pipeline that runs an existing open source software application and generates an automatic build 4.5I: Proof your solution by performing a complete release from a change in code that generates corresponding executables executing all the steps of a release management cycle Guarantee software quality by enabling quality tools and executing unit tests					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	40%	5.5	S1.05 & S1.07
TOETS02 (VT)	(Workplace) Assessment	Assessment	60%	5.5	S1.17 or S1.18 & S1.19

Block / Semester: J	
CU75082V1	**Title: Personal Professional Development Advanced**
Course information	
Amount of study credits: 7.5	**Language:** Dutch English
Conditions for course participation: none	
Conditions for test participation: none	
Brief description of course content: General bachelor competences in Agile working project groups and via CRISP-DM working project groups(by retrospective feedback or self-study). In this case: working in a planned manner, showing and coordinating appropriate efforts, motivated cooperation, team-oriented and self-managing actions, self-directed (team) learning, methodical judgments, communicative behaviour in a project context. Project management: the student learns the relationship of project management (PM) to software development, and concrete project management methods and methodologies are treated, such as SCRUM Project management. Students apply their PM skills during the projects and show what they have learned by showing deliverables and approving these deliverables through peer feedback. Students learn to communicate effectively in English in an IT project environment. During the English semester, students can practice, receive feedback and demonstrate a sufficient level of reading, understanding, writing and presentation skills for practical professional situations.	
Course learning outcomes: **Personal leadership** 7.2O: You're considerate, see opportunities and seize them. You have a proactive attitude that you take initiative and feel responsible for what you do You can motivate yourself and others, you are willing to help others / support (individual and team). You can present yourself or a team, take others into your own development. 7.2Q: You study demonstrates considered, strengthens your own learning and can recognize a learning need in yourself and mating act, reflect, evaluate, and give active feedback questions. You recognize when you need help and do it then You can specify what type of professional you want to be and / or what type of positions you aspire, know your own strengths and weaknesses and can describe yourself well. **Interact purposefully** 7.3C: You focus on the various groups of stakeholders such as partners, interest groups, individual team members etc 7.3D: You focus on what you want to communicate and what purpose you choose the most appropriate form and while you perform this proactively You focus on your role in the context of the ICT job, you recognize these tasks and takes proactive. You dare others to speak (feedback) and is open to feedback. You are open to other opinions / views / arguments and see that as an enrichment. You consciously builds confidence in an interdisciplinary and intercultural cooperation context. **Organise in a future-oriented way** You give evidence that you are able to think ahead and plan ahead. You think methodically about the approach suitable for the assignment (identification of tasks, order of execution, proper prioritization) and how this contributes to the end result. You plan and monitors the time. You are cost conscious. You recognize opportunities and risks. You can thereby all time aware of agreements, legal regulations and ethical standards. You have a keen eye for the feasibility of duties in the organization. You taking into account the characteristics of the area of the assignment. 7.4O: You examine where necessary and relevant to the ethical implications of the tasks you perform. You recognize their own and others' limits and act accordingly 7.4P: You can construct achievable and realistic goals within the time available which contribute to solving a problem or achieving a demand. The goals can be divided into multiple related detailed tasks **Solve problems in a research-oriented way** You provide evidence that your problems / challenges to identify and put in context (department / organization / business environment, social environment) and can analyse these problems. You are able, where appropriate and relevant to search for multiple solutions. 7.5E: You are curious Throughout the dissolution process, ask yourself if from different perspectives. You are pragmatically, creatively and critically and make if appropriate use of resources. You can make a thoughtful and methodical choosing the correct / most appropriate / suitable solution or approach.	

While you are critical about your own basis and used arguments.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Criterion-referenced interview	Criterium focused interview	100%	5.5	S2.11 to S2.15

Block / Semester: B2					
CU75078V1	**Title: User Value Creation**				
Course information					
Amount of study credits: 5			**Language:** English Dutch		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: User centred approach on creating a solution for a complex problem.					
Course learning outcomes: You describe the important consequences for UX based on a target group analysis 1.3L: You write a UX report accounting for design choices based on guidelines, human factors and/or emotional design 1.3M: You test the UX of a product in a UX test report to evaluate the quality 1.3N: You recommend further development steps based on the UX test report 4.2J: you describe the needs of the users of the software system to be developed you draw up a functional design for a complex part of a software system you write a techspecs report as reference that can be transferred to third parties You can implement a component for a given architecture					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Criterion-referenced interview	Group assessment based on products in a portfolio	100%	5.5	S1.19 & S1.20

Block / Semester: S1	
EN39301V26	**Title: Reading and Writing skills for IT**

Course information

Amount of study credits:	**Language:**
5	Dutch English

Conditions for course participation:

An overall B2 level IELTS/Cambridge or completion of EN39002/EN39003/EN39004, based on the results of the placement test.

Conditions for test participation:

Geen

Brief description of course content:

B2+ course.

The course is designed for second-year HBO-ICT students to address the critical need for improved reading comprehension and professional writing skills in the IT field. The course will be offered over two blocks, focusing on reading and writing. Students will enhance their ability to understand technical documents, infer tasks from instructions, and effectively communicate technical information through clear, structured documentation.

In the reading block, students will develop basic English reading comprehension relevant to the IT profession, understand text structures, and improve reasoning skills. The writing block will teach students to write for various audiences, communicate ideas effectively, and structure documents and narratives coherently. This course aims to provide students with transferrable skills that will benefit their academic and professional IT careers.

Course learning outcomes:

(referencing CEFR Companion Volume, 2018)

Reading Block

Comprehension (B2+)

- Can read with a large degree of independence, adapting style and speed of reading to different texts and purposes, and using appropriate reference sources selectively (B2, 60)
- Can understand a wide range of demanding, longer texts, and recognize implicit meaning (C1, 60)

Text Analysis (B2+)

- Can understand in detail lengthy, complex texts, provided he/she can reread difficult sections. (C1, 60)

Writing Block

Technical Writing (B2+)

- Can write clear, detailed texts on a variety of subjects related to their field of interest, synthesizing and evaluating information and arguments from a number of sources (B2, 75)
- Can express themselves in clear, well-structured text, expressing points of view at some length (C1, 75)

Document Structuring (B2+)

- Can write an essay or report that develops an argument systematically with appropriate highlighting of significant points and relevant supporting detail (B2, 77)
- Can, when preparing a more formal spoken or written text, consciously adopt the conventions linked to the particular type of text concerned (structure, level of formality and other conventions) (C1, 78)

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Written knowledge test	Reading test	50%	5.5	B2.8; B3.10
TOETS02 (VT)	Written knowledge test	Written test	50%	5.5	B3.8; B4.10

Block / Semester: B1	
EN39002	**Title: Foundation Course B2**

Course information

Amount of study credits: 5	**Language:** English

Conditions for course participation:
None

Conditions for test participation:
None

Assessability criteria:
None

Brief description of course content:
Students can take the placement test and/or consult the LCC teacher before they decide for which English foundation course they will register. Course level: B1/low B2 aiming at strong B2.
Learning Outcomes:

- ***Reading/ Use of English.*** Ability to: scan texts for relevant information; understand the gist of information and articles on nonfamiliar topics and understand most of the content; apply and adapt language suitable for B2.
- ***Writing***. Ability to: express opinions and give reasons; write a simple piece of academic writing (e.g. a report) giving some evaluation, advice etc.; present arguments using a limited range of vocabulary and grammatical structures.
- ***Listening***. Ability to: follow a talk or lecture on a familiar topic; keep up with conversations on a wide range of topics; understand the answers to factual questions asked.
- ***Speaking***. Ability to: ask for clarification and further information; check for understanding; express opinions and arguments to a limited extent; answer predictable and factual questions.

Based on CEFR. For more details see: https://www.coe.int/en/web/common-european-framework-reference-languages/level-descriptions

Note: students of IB - the minimum requirement is a C1 level

Course learning outcomes:
Strong B2 Level

Compulsory literature:
Open World B2, Anthony Cosgrove, Deborah Hobbs, 1st edition, ISBN: 9781108759052

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Written knowledge test	Reading and Use of English	40%	5.5	B1.8; B2.8; B3.8; B3.10; B4.8; B4.10
TOETS02 (VT)	Written knowledge test	Writing	20%	5.5	B1.8; B2.8; B3.8; B3.10; B4.8; B4.10
TOETS03 (VT)	Written knowledge test	Listening	20%	5.5	B1.8; B2.8; B3.8; B3.10; B4.8; B4.10
TOETS04 (VT)	Assignment (group)	Speaking	20%	5.5	B2.2; B2.7; B2.9; B3.2; B3.7; B3.9; B4.2; B4.7; B4.9; B4.10

Block / Semester: B1	
EN39003	**Title: Foundation Course C1**

Course information

Amount of study credits: 5	**Language:** English

Conditions for course participation:
None

Conditions for test participation:
None

Assessability criteria:
None

Brief description of course content:
Students can take the placement test and/or consult the LCC teacher before they decide for which English foundation course they will register. Course Level: B2/low C1 aiming at strong C1
Learning Outcomes:

- ***Reading/Use of English.*** Ability to: read quickly enough to cope with an academic course delivered in English; understand complex and arguments in lectures without serious misunderstandings; scan texts for relevant information and understand the gist of the text; apply and adapt language suitable for C1.
- ***Writing***. Ability to: make reasonable accurate notes in meetings and lectures; write a piece of work whose message can be followed throughout; write a piece of work showing the ability to communicate with no serious errors.
- ***Listening and speaking***. Ability to: contribute effectively in meetings and seminars in own field of study, probing for more information if required; maintain a casual conversation with a good degree of fluency; take part in an abstract conversation with a good degree of fluency; follow discussions and arguments with only occasional need for clarification; employ good compensation strategies to overcome linguistic inadequacies; deal with unpredictable questions; give critical feedback in a non-offensive manner.

Based on CEFR. For more details see: https://www.coe.int/en/web/common-european-framework-reference-languages/level-descriptions

Note: students of IB - the minimum requirement is a C1 level

Course learning outcomes:
Strong C-1 level

Compulsory literature:
Open World C1 Advanced Student's Book with Answers, Anthony Cosgrove, Claire Wijayatilake, 1st edition, ISBN: 9781108891455

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Written knowledge test	Reading and Use of English	40%	5.5	B1.8; B2.8; B3.8; B3.10; B4.8; B4.10
TOETS02 (VT)	Written knowledge test	Writing	20%	5.5	B1.8; B2.8; B3.8; B3.10; B4.8; B4.10
TOETS03 (VT)	Written knowledge test	Listening	20%	5.5	B1.8; B2.8; B3.8; B3.10; B4.8; B4.10
TOETS04 (VT)	Assignment (group)	Speaking	20%	5.5	B2.2; B2.7; B2.9; B3.2; B3.7; B3.9; B4.2; B4.7; B4.9; B4.10

Block / Semester: B1	
EN39004	**Title: Foundation Course C2**

Course information

Amount of study credits: 5	**Language:** English

Conditions for course participation:
None

Conditions for test participation:
None

Assessability criteria:
None

Brief description of course content:
Students can take the placement test and/or consult the LCC teacher before they decide for which English foundation course they will register. Course level: C1/low C2 aiming at strong C2.
Learning Outcomes:

- ***Reading/Use of English***. Ability to: understand complex documents and reports; understand academic articles in a relevant field including complex ideas expressed in complex language; access all sources of information quickly and reliably; apply and adapt language suitable for C2.
- ***Writing***. Ability to: make full notes of meetings and seminars with good expression and accuracy; make full notes of meetings and seminars while continuing to participate; make accurate and complete notes of a lecture.
- ***Listening and speaking***. Ability to: advise on or talk about sensitive or complex issues (within field of knowledge) with ease; deal confidently with hostile questions; speak fluently and express/understand nuances of language; present a clear, smooth-flowing description or argument in a style appropriate to the context with an effective logical structure.

Based on CEFR. For more details see: https://www.coe.int/en/web/common-european-framework-reference-languages/level-descriptions

Note: students of IB - the minimum requirement is a C1 level

Course learning outcomes:
Strong C-2 level

Compulsory literature:
Objective Proficiency Student's Book with Answers with Downloadable Software, Annette Capel and Wendy Sharp, ISBN: 9781107646377

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Written knowledge test	Reading and Use of English	40%	5.5	B1.8; B2.8; B3.8; B3.10; B4.8; B4.10
TOETS02 (VT)	Written knowledge test	Writing	20%	5.5	B1.8; B2.8; B3.8; B3.10; B4.8; B4.10
TOETS03 (VT)	Written knowledge test	Listening	20%	5.5	B1.8; B2.8; B3.8; B3.10; B4.8; B4.10
TOETS04 (VT)	Assignment (group)	Speaking	20%	5.5	B2.2; B2.7; B2.9; B3.2; B3.7; B3.9; B4.2; B4.7; B4.9; B4.10

Block / Semester: B3	
CU75073V3	**Title: Data Driven Decision making**

Course information

Amount of study credits:	**Language:**
10	Dutch English

Conditions for course participation:
none

Conditions for test participation:

Brief description of course content:
Getting acquainted with the iterative Data Science process, in which all the stages of the cycle are completed. The emphasis is on creating data-based insights into complex issues. Students work in teams with CRISP-DM methodology on a Data Science project in mixed NL and ENG groups.
The student still works in sprints but follows the steps of CRISP-DM. Python classes are introduced to educate the much-needed skill set in data science projects. Deliverables are delivered to the client in a demo, and the steps are evaluated. Deliverables are delivered in a professional portfolio.
The first steps are business and data understanding. Therefore, students analyse the organisation, including organisational processes, using standardised methods. Organisational analysis and the first phases of CRISP-DM are combined end the deliverables are delivered in a professional portfolio. Further, students learn to be able to view systems, data and IT solutions from a security perspective. Estimating the impact of data, software and IT-related developments on society from an ethical perspective and elaborating on different points of view.

Course learning outcomes:

Test 1:
2.1B: analyse the performance of an organization through a standard methodology
2.1C: map an organization process of an existing organization by using suitable methodologies
2.1D: you assess a given situation on various security aspects
2.1I: you map the branch and the company and you analyse how that process contributes to the company's goals
2.4A: you submit a sound analysis report based on a company organization analysis

Test 2:
Learning Outcome 1: Focus on Business Understanding: You setup a data science process (learning goals: 6.1A, 6.1B, 6.1C, 6.1D, 6.1E)
Learning Outcome 2: Focus on Data Understading & Data Preperation: You collect and address relevant data (learning goals: 6.2A, 6.2B, 6.2C, 6.2D, 6.2E, 6.2F)
Learning Outcome 3: Focus on Modeling: You perform data analysis (learning goals: 6.3A, 6.3B, 6.3C)
Learning Outcome 4: Focus on Evaluation: You evaluate & deploy results of the data science process (learning goals: 6.4A, 6.4B)

T**est 3:**
6.4C: You produce a final report and present this to customer
6.4D: You review the data science process and you determine, and also report, lessons learned
7.3Q: as a team you can communicate your research in an organized way, appropriate for the audience
7.3P: Students can present their project, the content of their portfolio and their process considerations in a sound way making plausible the equal contribution of each project member to the project

Test 4
7.2D: You are able to assess an ethical situation, following the steps of an ethical framework, and incorporate different schools of thought such as duty, virtue, and consequentialism ethics, in sound reasoning, i.e. avoiding logical fallacies
4.5K: You master the basic concepts of (symmetrical) encryption, and argue for a given situation, how symmetrical encryption can improve the security of data.

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)

TOETS01 (VT)	Portfolio	Portfolio (group)	30%	5.5	S2.7 & S2.9
TOETS02 (VT)	Written knowledge test	Quiz (individual)	30%	5.5	S2.9 & S2.10
TOETS03 (VT)	Assignment (group)	Assingment (portfolio)	15%	5.5	S2.8 & S2.10
TOETS04 (VT)	Written knowledge test	Knowledge test (individual)	25%	5.5	S2.5 & S2.10

Block / Semester: B4	
CU75074V3	**Title: Data Science/AI**
Course information	
Amount of study credits: 7.5	**Language:** Dutch English
Conditions for course participation: none	
Conditions for test participation:	
Brief description of course content: Getting acquainted with the iterative Data Science process, in which all the stages of the cycle are completed. The emphasis is on creating insight, based on data, for complex issues. Students work in teams with CRISP-DM methodology on a Data Science project in mixed NL and ENG groups. Students still work in sprints but follow the steps of CRISP-DM. Python classes are introduced to educate the much-needed skill set in data science projects. Deliverables are delivered to the client. Deliverables are delivered in a professional manner. The first steps are business and data understanding. Therefore, students analyse the organisation, including organisational processes, using standardised methods. Organisational analysis and the first phases of CRISP-DM are combined, and the deliverables are delivered in a professional format. **@TODO: CRISP-DM terminologie toelichten**	
Course learning outcomes: **Test 1:** **Learning Outcome 1: Focus on Business Understanding: You setup a data science process (learning goals: 6.1F, 6.1G, 6.1H)** **Learning Outcome 2: Focus on Data Understading & Data Preperation: You collect and address relevant data (learning goals: 6.2G, 6.2H, 6.2I, 6.2J, 6.2K)** **Learning Outcome 3: Focus on Modeling: You perform data analysis (learning goals: 6.3D, 6.3E, 6.3F)** **Learning Outcome 4: Focus on Evaluation: You evaluate & deploy results of the data science process (learning goals: 6.4E, 6.4F, 6.4G, 6.4H)** 6.1F: You define data mining goals success criteria 6.1G:You describe data mining activities based on choice of a basic machine learning model and relevant required activities 6.1H: You add extra self-organised and/or external data sources to the data science process 6.2G:You (re-)validate data after model generated assumptions 6.2H:You clean data by imputating and scaling relevant data 6.2I: You construct data by one-hot-encoding, defining targets & labelling relevant data 6.2J:You integrate relevant data by merging multiple data sources 6.2K:You convert data formats as prerequisite for relevant model(s) 6.3D: You split data into test & train sets to generate a test design 6.3E:You build & train relevant model(s) and create predictions using the model(s) on test data set 6.3F:You assess the model(s) on chosen metrics of the defined success criteria 6.4E:You evaluate and match success criteria with business objectives of the data science process 6.4F:You determine next steps and setup an advisory report for follow-up 6.4G:You produce a deliverable for customer 6.4H:You review the data science process and collect lessons learned on process & product **Test 2:** 7.3Q: as a team you can communicate your research in an organized way, appropriate for the audience. 7.3P: Students can present their project, the content of their portfolio and their process considerations in a sound way making plausible the equal contribution of each project member to the project	

7.3R: Students are able to deliver a solid product demonstration to the stakeholders in which they demonstrate the product and address the main challenges and present a realistic roadmap					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Written knowledge test	Quiz (individual)	60%	5.5	S2.19 & S2.20
TOETS02 (VT)	Assignment (group)	Project (group)	40%	5.5	S2.18 & S2.20

Block / Semester: B2	
CU75020V3	**Title: Software Design**

Course information

Amount of study credits:	**Language:**
5	Dutch English

Conditions for course participation:
none

Conditions for test participation:
none

Brief description of course content:
Make software robust! Learn how to detect weak spots in programming code (code smells) and how to solve them (refactoring) with proven solutions like design patterns. Student will learn to Detect design patterns with a tool in an open source software system and will report the result (including class diagram) in a short report. Student will learn to apply refactoring in an open source software system and report their findings and opinion in a blog. Students will Create in pairs a working program that houses multiple design patterns.

Course learning outcomes:

Test 1:
4.3J: Indicate for a given code example/class diagram which design patterns were applied
4.3L: Recognise weak points in code, so-called code smells, and apply an appropriate standardised remedy, so-called refactoring

Test 2:
4.3K: Apply a suitable design pattern for a given situation and work it out in both a class diagram and actual code

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Report (individual)	50%	5.5	S1.15 & S1.19
TOETS02 (VT)	Assignment (individual)	Program (group)	50%	5.5	S1.17 & S1.20

Block / Semester: B1 **Block / Semester: B4**	
CU75028V2	**Title: Cloud Computing**

Course information

Amount of study credits: 5	**Language:** English Dutch

Conditions for course participation:
none

Conditions for test participation:
none

Brief description of course content:
Use cloud specific building blocks like serverless functions and different kinds of cloud storage, learn how to connect and monitor them, to let your project scale on a new level.
Course DVI is mandatory for study track Business IT Consultant.
Course CCO is mandatory for study track Software Engineer.
Course CCO & DVI are mandatory for study track Data Science. DVI will take place in year 2 and CCO in year 4.

Course learning outcomes:
Make available a software system based on a Framework for users in a simple hosting environment
3.4A: The student can advise for a given project how it should be adapted to be able to use the functionalities of a cloud provider
3.5A: The student can select and employ and react accordingly to the generated metrics for cloud application control tools.

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Research proposal	40%	5.5	S1.3 & S1.9 OR S2.13 & S2.19
TOETS02 (VT)	Assignment (individual)	Research report and proof of concept	60%	5.5	S1.7 & S1.9 OR S2.17 & S2.19

Block / Semester: B2					
CU75072V2	**Title: Data Driven Business**				
Course information					
Amount of study credits: 5			**Language:** English Dutch		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: Introduction in “how to become a data driven organization”. Students will learn the definition of Data Driven business and why companies want or need to change their business. Students are given tools to determine which companies are data driven. Furthermore they will have understanding in what is needed for companies to become data driven. Additionally, from a maturity point of view, students will be introduced to an exemplary roadmap in which a company may become data driven. In addition, students are given insight in flaws, failures & don’ts of becoming data driven. All aspects of the courses will be backed by real-life cases, so far as possible. Lastly the connection to Data Strategy will be explained, to ensure students understand what the end-goals may look like in a broader overview. Students will work in groups of 3 or 4 (depends on the number of students starting the course).					
Course learning outcomes: You are capable of understanding the need for business to embrace data and can report what their maturity in this field is You understand how a company’s data maturity fits in a broader context of data strategy 2.4F: You can advise about the future perspective of data driven business.					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio (group)	50%	5.5	S1.18 & S1.20
TOETS02 (VT)	Presentation (individual)	Presentation (group)	50%	5.5	S1.18 & S1.20

Block / Semester: B4					
CU75027V3	**Title: Data Visualisation**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: Creating a suitable data visualization for communicating information to a client. You will learn about data visualization goals, types and characteristics and how to research the best choices for your specific case. To conclude your research, you will create an actual data visualization (proof of concept). Course DVI is mandatory for study track Business IT Consultant. Course CCO is mandatory for study track Software Engineer. Course CCO & DVI are mandatory for study track Data Science. DVI will take place in year 2 and CCO in year 4.					
Course learning outcomes: 1.4A: You can draw up a datavision goal based on the project context and business goal taking into account the goal, the target group and the message You can make a sound choice for a datavisualisation type suitable for the datavisualisation goal You can make a sound choice for visual elements suitable for the datavisualisation goal 1.4D: You can realise a datavisualisation based on sound research					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assingment (individual)	100%	5.5	S2.17 & S2.19

Block / Semester: J					
CU75058V1	**Title: IT Personality Projectweek 2**				
Course information					
Amount of study credits: 1.25			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: This course can be followed 3 times during the study programme. Course description for CU75054, CU75058, and CU75075 are identical. IT Personality content is based on the HZ-wide programme HZ personality that stimulates the skills concerning and attitudes towards personal development and personal leadership. The programme can either have a broadening or a deepening focus when it comes to the curriculum. Each year the ICT program organizes a projectweek with real life casus and (if possible) in cooperation with other programs. This projectweek course can be chosen as 1,25 ec content for personality. The assessment criteria and assessment process are listed in the IT Personality 2021-2022 instruction manual which can be found on the Learn page. This course is already approved for IT personality, students only need to define their personal goals within the given context.					
Course learning outcomes: 7.2M: Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals through participation in a project week					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	-	See Learn manual

Block / Semester: J					
CU75075V1	**Title: IT Personality Projectweek 3**				
Course information					
Amount of study credits: 1.25			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: This course can be followed 3 times during the study programme. Course description for CU75054, CU75058, and CU75075 are identical. IT Personality content is based on the HZ-wide programme HZ personality that stimulates the skills concerning and attitudes towards personal development and personal leadership. The programme can either have a broadening or a deepening focus when it comes to the curriculum. Each year the ICT program organizes a projectweek with real life casus and (if possible) in cooperation with other programs. This projectweek course can be chosen as 1,25 ec content for personality. The assessment criteria and assessment process are listed in the IT Personality 2021-2022 instruction manual which can be found on the Learn page. This course is already approved for IT personality, students only need to define their personal goals within the given context.					
Course learning outcomes: 7.2M: Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals through participation in a project week					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	-	See learn manual

Block / Semester: J					
CU75059V1	**Title: IT Personality 3**				
Course information					
Amount of study credits: 1.25			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: IT Personality content is based on the HZ-wide programme HZ personality that stimulates the skills concerning and attitudes towards personal development and personal leadership. The programme can either have a broadening or a deepening focus when it comes to the curriculum. A prerequisite for starting the HZ Personality related activities is having obtained a GO from one of the IT personality coordinators. The assessment criteria and assessment process are listed in the HZ Personality 2021-2022 instruction manual which can be found on the Learn page.					
Course learning outcomes: 7.2L: Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals, community goals and personal goals					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	-	See Learn manual

Block / Semester: J					
CU75060V1	**Title: IT Personality 4**				
Course information					
Amount of study credits: 1.25			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: IT Personality content is based on the HZ-wide programme HZ personality that stimulates the skills concerning and attitudes towards personal development and personal leadership. The programme can either have a broadening or a deepening focus when it comes to the curriculum. A prerequisite for starting the HZ Personality related activities is having obtained a GO from one of the IT personality coordinators. The assessment criteria and assessment process are listed in the HZ Personality 2021-2022 instruction manual which can be found on the Learn page.					
Course learning outcomes: 7.2L: Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals, community goals and personal goals					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	-	See Learn manual

Block / Semester: J					
CU75061V1	**Title: IT Personality 5**				
Course information					
Amount of study credits: 1.25			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: IT Personality content is based on the HZ-wide programme HZ personality that stimulates the skills concerning and attitudes towards personal development and personal leadership. The programme can either have a broadening or a deepening focus when it comes to the curriculum. A prerequisite for starting the HZ Personality related activities is having obtained a GO from one of the IT personality coordinators. The assessment criteria and assessment process are listed in the HZ Personality 2021-2022 instruction manual which can be found on the Learn page					
Course learning outcomes: 7.2L: Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals, community goals and personal goals					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	-	See Learn manual

Block / Semester: J					
CU75062V1	**Title: IT Personality 6**				
Course information					
Amount of study credits: 1.25			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: IT Personality content is based on the HZ-wide programme HZ personality that stimulates the skills concerning and attitudes towards personal development and personal leadership. The programme can either have a broadening or a deepening focus when it comes to the curriculum. A prerequisite for starting the HZ Personality related activities is having obtained a GO from one of the IT personality coordinators. The assessment criteria and assessment process are listed in the HZ Personality 2021-2022 instruction manual which can be found on the Learn page.					
Course learning outcomes: 7.2L: Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals, community goals and personal goals					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	-	See Learn manual

Block / Semester: J					
CU75063V1	**Title: IT Personality 7**				
Course information					
Amount of study credits: 1.25			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: IT Personality content is based on the HZ-wide programme HZ personality that stimulates the skills concerning and attitudes towards personal development and personal leadership. The programme can either have a broadening or a deepening focus when it comes to the curriculum. A prerequisite for starting the HZ Personality related activities is having obtained a GO from one of the IT personality coordinators. The assessment criteria and assessment process are listed in the HZ Personality 2021-2022 instruction manual which can be found on the Learn page.					
Course learning outcomes: 7.2L: Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals, community goals and personal goals					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	-	See Learn manual

Block / Semester: J					
CU75064V1	**Title: IT Personality 8**				
Course information					
Amount of study credits: 1.25			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: IT Personality content is based on the HZ-wide programme HZ personality that stimulates the skills concerning and attitudes towards personal development and personal leadership. The programme can either have a broadening or a deepening focus when it comes to the curriculum. A prerequisite for starting the HZ Personality related activities is having obtained a GO from one of the IT personality coordinators. The assessment criteria and assessment process are listed in the HZ Personality 2021-2022 instruction manual which can be found on the Learn page.					
Course learning outcomes: 7.2L: Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals, community goals and personal goals					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	-	See Learn manual

Block / Semester: J					
CU75065V1	**Title: IT Personality 9**				
Course information					
Amount of study credits: 1.25			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: IT Personality content is based on the HZ-wide programme HZ personality that stimulates the skills concerning and attitudes towards personal development and personal leadership. The programme can either have a broadening or a deepening focus when it comes to the curriculum. A prerequisite for starting the HZ Personality related activities is having obtained a GO from one of the IT personality coordinators. The assessment criteria and assessment process are listed in the HZ Personality 2021-2022 instruction manual which can be found on the Learn page.					
Course learning outcomes: 7.2L: Developing skills and behavior to achieve personal and professional goals. Carrying out activities that contribute to sustainable development goals, community goals and personal goals					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	-	See Learn manual

Block / Semester: J	
CU75033V2	**Title: Internship SE**

Course information

Amount of study credits:	**Language:**
25	Dutch English

Conditions for course participation:

- the student is in possession of the propaedeutic certificate of the HBO-ICT program;
- the student has obtained at least 30 EC of completed courses in the second year of the program (semesters 3 and 4)

Conditions for test participation:

none

Brief description of course content:

The internship of the HBO-ICT program aims to: learn to function professionally in a business, ICT-related environment. This is achieved by the student by setting his own learning objectives based on the HBO-ICT professional competences and by reflecting on his own performance. It concerns primarily professional tasks specifically in the field of software engineering.

Course learning outcomes:

4.1N, You can independently make an analysis of a software engineering design problem in an internship context
4.2P, You can independently select, document, communicate and evaluate solutions for a software engineering design problem in an internship context using tests and prototypes
4.3O, You can independently realise a suitable solution for a software engineering design problem in an internship context
4.4A, You can independently give a suitable advice for solving a software engineering design problem in an internship context
7.1Q: You can function professionally in a company-related, ICT-related environment

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	5.5	B1: S1.18 & S1.20 or S2.8. B2: S2.8 & S2.10 or S2.18 B3: S2.18 & S2.20 or S1.8

Block / Semester: J	
CU75034V2	**Title: Internship DS**

Course information

Amount of study credits:	**Language:**
25	Dutch English

Conditions for course participation:

- the student is in possession of the propaedeutic certificate of the HBO-ICT program;
- the student has obtained at least 30 EC of completed courses in the second year of the program (semesters 3 and 4)

Conditions for test participation:

none

Brief description of course content:

The internship of the HBO-ICT program aims to: learn to function professionally in a business, ICT-related environment. This is achieved by the student by setting his own learning objectives based on the HBO-ICT professional competences and by reflecting on his own performance. It concerns primarily professional tasks specifically in the field of software engineering.

Course learning outcomes:

6.1K: You can independently set up a data science process in a internship context
6.2Q: You independently collect and address relevant data in a internship context
6.3J: You can independently perform data analysis in a internship context
6.4K: You can independently evaluate and deploy results of a data science process in a internship context
7.1Q: You can function professionally in a company-related, ICT-related environment

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	5.5	B1: S1.18 & S1.20 or S2.8. B2: S2.8 & S2.10 or S2.18 B3: S2.18 & S2.20 or S1.8

Block / Semester: J					
CU75035V2	**Title: Internship BIC**				
Course information					
Amount of study credits: 25			**Language:** Dutch English		
Conditions for course participation: • the student is in possession of the propaedeutic certificate of the HBO-ICT program; • the student has obtained at least 30 EC of completed courses in the second year of the program (semesters 3 and 4)					
Conditions for test participation: none					
Brief description of course content: The internship of the HBO-ICT program aims to: learn to function professionally in a business, ICT-related environment. This is achieved by the student by setting his own learning objectives based on the HBO-ICT professional competences and by reflecting on his own performance. It concerns primarily professional tasks specifically in the field of software engineering.					
Course learning outcomes: 2.1G: You can independently make a validated "process" analysis for an ICT provision in the context of an internship 2.2D: You can independently make a validated process design and understand the relationship with the information provision in the context of an internship. 2.5A: You can independently draw up a management plan for ICT processes in a internship context according to a chosen framework, taking into account updating, design, maintenance and quality assurance 7.1Q: You can function professionally in a company-related, ICT-related environment					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	5.5	B1: S1.18 & S1.20 or S2.8. B2: S2.8 & S2.10 or S2.18 B3: S2.18 & S2.20 or S1.8

Block / Semester: J					
CU75048V3	**Title: Graduation Preparation**				
Course information					
Amount of study credits: 5			**Language:** English Dutch		
Conditions for course participation: none					
Conditions for test participation: The student is allowed to take the test when allowed course participation Complex project* * BIC = CU75066V1, SE = CU75047V2, DS = CU75067V1					
Brief description of course content: In this course the student will be prepared on their graduation. This includes workshops about the transition from student to professional but also guidance on finding a graduation company that is a good fit to the student, guidance in writing a graduation proposal and guidance in writing a graduation plan including research related tools.					
Course learning outcomes: 7.5A: you can make a proposal for a sufficiently complex graduation assignment 7.5B: you can draw up a graduation plan for a complex graduation assignment					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Report	Assignment (individual)	100%	5.5	S1.7, S1.17, S2.7, S2.17

Block / Semester: B1					
CU75045V1	**Title: Modern Programming Practices**				
Course information					
Amount of study credits: 5			**Language:** English Dutch		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: Discuss and apply advanced programming and software engineering concepts, such as concurrency and paradigms, while taking into account the impact on quality of the software.					
Course learning outcomes: you recognise and explain with which programming techniques you can solve certain software problems 4.3M: you apply the right combination of programming techniques for the problems in a complex software system					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Assignment (individual)	Assignment (individual)	100%	5.5	S1.8 or S1.9 & S1.10

Block / Semester: B2					
CU75042V2	**Title: Web Application Security**				
Course information					
Amount of study credits: 5			**Language:** English Dutch		
Conditions for course participation: None					
Conditions for test participation: • The student is in possession of the propaedeutic certificate of the HBO-ICT program; • The student has obtained at least 60 EC from the main phase with completed courses; • The student has passed the internship					
Brief description of course content: In terms of content, various (advanced) security principles are cited in this course. Students must map and audit an existing, complex system using the given methods and then write appropriate advice about security in the system.					
Course learning outcomes: you can map the trust boundaries of a complex system you perform a security audit through a given model 4.4B: you write a suitable advice on the results of a security research that was held you explain the results of the security audit according to a model					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Advice report	100%	5.5	S1.16 & S1.19

Block / Semester: S1	
CU75047V3	**Title: Complex Project SE**

Course information

Amount of study credits:	**Language:**
15	Dutch English

Conditions for course participation:

- The student is in possession of the propaedeutic certificate of the HBO-ICT programme;
- The student has obtained at least 60 EC from the main phase with completed courses;
- The student has passed the internship (CU75033)

Conditions for test participation:

- The student is in possession of the propaedeutic certificate of the HBO-ICT programme;
- The student has obtained at least 60 EC from the main phase with completed courses;
- The student has passed the internship (CU75033)

Brief description of course content:

In this course the student will do a complex project in a small group under coaching of lecturers and experts. The project and professional products will be specific for the study track. The form and account of the results are similar with the graduation phase.

Course learning outcomes:

Test 1:

4.1O: you describe functional and quality specifications and limiting preconditions, in which at least maintenance and manageability are included in the local infrastructure and development processes
4.1P: you use various types of sources and techniques for collecting specifications and preconditions.
4.1Q: You can validate the formulated specifications and preconditions and thus assess the degree of completeness and objectivity
4.1R: You can thoroughly describe a technical and/or process-related problem concerning the production of software
4.2Q: you evaluate solutions based on the stated specifications and limitations (consistency) using tests, prototypes and comparable techniques. In addition, you analyse data collected with qualitative and/or quantitative analysis techniques
4.2R: you select candidate solutions based on relevant, current and specialist professional knowledge from the ICT domain
4.2S: you apply appropriate schematic techniques in the document where possible, which are in line with the chosen design strategy and goaled at the target group, which in any case consists of developers who (further) develop the product.
4.3P: you realise (prototypes of) a system existing of several sub systems and/or existing components
4.3Q: You can do research into the quality of the realised software such as functionality, security and performance
4.4D: you advise the customer on a solution for a software problem, convince the customer that the solution is in line with his/her objective and vision and you support the customer in the implementation of the solution or you give you process-oriented advice

Test 2:

7.1M: you can employ the right professional skills to complete a project successfully in a complex environment
7.1N: you account for the choices made regarding the professional skills employed
7.3M: As a project group you deliver structured products and account for everyone's role within the project, the method followed and evaluate the process and the product critically

Test 3:

7.3L: As a project group you can report and present professionally, both verbally and in a report

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Groupassignment	60%	5.5	S2.18 & S2.20
TOETS02 (VT)	Portfolio	Portfolio (individual)	20%	5.5	S2.16 & S2.19

TOETS03 (VT)	Criterion-referenced interview	Criterium focused interview	20%	5.5	S2.19 & S2.20

Block / Semester: J	
CU75050V1	**Title: Graduation Software Engineering**

Course information

Amount of study credits:	**Language:**
30	English Dutch

Conditions for course participation:

- The student is in possession of the propaedeutic certificate of the HBO-ICT program;
- The student has obtained at least 137.5 EC from the main phase with completed courses.

Conditions for test participation:

As included in Graduation Student Manual on the graduation Learn page

Brief description of course content:

Students conduct their graduation on a complex practical assignment in a complex situation. The students does this independently. The final products are qualitative sufficient professional software engineering products, supplemented with an account of the methodical and professional approach.
Final results will be presented followed by an assessment of two examiners and possibly one external expert.

Course learning outcomes:

Test 1:

you can independently in a complex environment employ the right professional skills to complete a project successfully
7.1P: you account for the choices made regarding the professional skills employed
7.3O: You deliver structured products, account for the method followed and evaluate the process and the product critically
7.3P: Students can present their project, the content of their portfolio and their process considerations in a sound way making plausible the equal contribution of each project member to the project.
You independently make an analysis of a software engineering design problem in a complex context
4.2T: You can independently select, evaluate (partial), document and communicate solutions for a software engineering design problem in a complex context
4.3R: You independently realise a suitable solution to a software engineering design problem in a complex context, independently
4.4E: You independently give a suitable advice for solving a software engineering design problem in a complex context

Test 2:

7.3N: You can report and present professionally, both verbally and in a report

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio+assessment	80%	5.5	S1.7, S1.10, S1.17, S1.20, S2.7, S2.10, S2.17, S2.20
TOETS02 (VT)	Criterion-referenced interview	Criterium focused interview	20%	5.5	S1.9, S1.10, S1.19, S1.20, S2.9, S2.10, S2.19, S2.20

Block / Semester: B2					
CU75046V2	**Title: Data Management & Governance**				
Course information					
Amount of study credits: 5			**Language:** Dutch English		
Conditions for course participation: none					
Conditions for test participation: none					
Brief description of course content: In this course you will get to know all aspects related to data management based on the DM-Boks. In addition, a number of aspects are chosen that are deepened (think of legal aspects, GDPR, Meta data etc.)					
Course learning outcomes: 6.1L: You can compose a data management plan for a specific project, taking in account al facets of a given, recognised standard					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	5.5	S1.18 & S.1.20

Block / Semester: S1	
CU75067V2	**Title: Complex Project DS**

Course information

Amount of study credits:	**Language:**
15	Dutch English

Conditions for course participation:

- The student is in possession of the propaedeutic certificate of the HBO-ICT program;
- The student has obtained at least 60 EC from the main phase with completed courses;
- The student has passed the internship (CU75034)

Conditions for test participation:

- The student is in possession of the propaedeutic certificate of the HBO-ICT program;
- The student has obtained at least 60 EC from the main phase with completed courses;
- The student has passed the internship (CU75034)

Brief description of course content:

In this course the student will do a complex project in a small group under coaching of lecturers and experts. The project and professional products will be specific for the study track. The form and account of the results are similar with the graduation phase.

Course learning outcomes:

Test 1:

Learning Outcome 1: Focus on Business Understanding: You setup a data science process (learning goals: 6.1J)
Learning Outcome 2: Focus on Data Understading & Data Preperation: You collect and address relevant data (learning goals: 6.2L, 6.2H, 6.2N, 6.2O, 6.2P)
Learning Outcome 3: Focus on Modeling: You perform data analysis (learning goals: 6.3G, 6.3H, 6.3I)
Learning Outcome 4: Focus on Evaluation: You evaluate & deploy results of the data science process (learning goals: 6.4I, 6.4J)
7.3M: As a project group you deliver structured products and account for everyone's role within the project, the method followed and evaluate the process and the product critically

Test 2:

7.1M: you can employ the right professional skills to complete a project successfully in a complex environment
7.1N: you account for the choices made regarding the professional skills employed

Test 3:

7.3L: As a project group you can report and present professionally, both verbally and in a report

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Groupassignment	60%	5.5	S2.18 & S2.20
TOETS02 (VT)	Portfolio	Portfolio (individual)	20%	5.5	S2.19 & S2.20
TOETS03 (VT)	Criterion-referenced interview	Criterium focused interview	20%	5.5	S2.16 & S2.19

Block / Semester: J	
CU75049V1	**Title: Graduation Data Science**

Course information

Amount of study credits:	**Language:**
30	English Dutch

Conditions for course participation:

- The student is in possession of the propaedeutic certificate of the HBO-ICT program;
- The student has obtained at least 137.5 EC from the main phase with completed courses.

Conditions for test participation:

As included in Graduation Student Manual on the graduation Learn page.

Brief description of course content:

Students conduct their graduation on a complex practical assignment in a complex situation. The students does this independently. The final products are qualitative sufficient professional software engineering products, supplemented with an account of the methodical and professional approach.
Final results will be presented followed by an assessment of two examiners and possibly one external expert.

Course learning outcomes:

Test 1:
7.1O: you can independently in a complex environment employ the right professional skills to complete a project successfully
7.1P: you account for the choices made regarding the professional skills employed.
7.3O: You deliver structured products, account for the method followed and evaluate the process and the product critically
6.1L: You can independently set up a data science process in a complex context.
6.2R: You independently collect and address relevant data in a complex context
6.3K: You can independently perform data analysis in a complex context.
6.4L: You can independently evaluate and deploy results of a data science process in a complex context

Test 2:
7.3N: You can report and present professionally, both verbally and in a report

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio+assessment	80%	5.5	S1.7, S1.10, S1.17, S1.20, S2.7, S2.10, S2.17, S2.20
TOETS02 (VT)	Criterion-referenced interview	Criterium focused interview	20%	5.5	S1.9, S1.10, S1.19, S1.20, S2.9, S2.10, S2.19, S2.20

Block / Semester: S1	
CU75044V2	**Title: Change, Yes you Can**

Course information

Amount of study credits:	**Language:**
5	English Dutch

Conditions for course participation:

- The student is in possession of the propaedeutic certificate
- The student has obtained at least 60 EC from the main phase with completed courses;
- The student has passed the internship

Conditions for test participation:

- The student is in possession of the propaedeutic certificate
- The student has obtained at least 60 EC from the main phase with completed courses;
- The student has passed the internship

Brief description of course content:

In terms of content, the soft skills in the field of conversation techniques are practiced in this course (how do you deal with a bad news conversation, how do you deal with resistance, how do you deal with someone who does not listen, etc.). The hard skills are researching change strategies, so that you can implement this theory later in the project.

Course learning outcomes:

Test 1:
7.3K: you can communicate in a sound way with various departments within a company, taking into account hierarchical layers

Test 2:
2.2B: you can map sound change strategies, so that you can choose the right strategy for the right change/company in a methodical way
You show by a case that you apply the theory into practice

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Criterion-referenced interview	Criterium based interview	50%	5.5	S1.7 & S1.9
TOETS02 (VT)	Portfolio	Portfolio	50%	5.5	S1.7 & S1.9

Block / Semester: S1					
CU75043V2	**Title: Making Business Intelligent**				
Course information					
Amount of study credits: 5			**Language:** English Dutch		
Conditions for course participation: • The student is in possession of the propaedeutic certificate; • The student has obtained at least 60 EC from the main phase with completed courses; • The student has passed the internship					
Conditions for test participation: • The student is in possession of the propaedeutic certificate; • The student has obtained at least 60 EC from the main phase with completed courses; • The student has passed the internship					
Brief description of course content: In terms of content, various (advanced) data sets are used in this course to ultimately display self-invented KPIs in a BI report.					
Course learning outcomes: you understand the importance of a sound BI report you understand what the necessity of BI is for companies 2.1L: You analyze and evaluate the impact of Business Intelligence (BI) on the architecture of an organization from the perspective of Enterprise Architecture. 2.2C: you understand the ETL and the matching report process 2.3A: you create KPIs for a dataset that you substantiate yourself and create a matching BI report 2.3B: you carry out the entire process from importing the data to creating the report					
Assessment information					
Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio	100%	5.5	S1.17 & S1.19

Block / Semester: S1	
CU75066V2	**Title: Complex Project BIC**

Course information

Amount of study credits:	**Language:**
15	English Dutch

Conditions for course participation:

- The student is in possession of the propaedeutic certificate of the HBO-ICT program;
- The student has obtained at least 60 EC from the main phase with completed courses;
- The student has passed the internship (CU75035)

Conditions for test participation:

- The student is in possession of the propaedeutic certificate of the HBO-ICT program;
- The student has obtained at least 60 EC from the main phase with completed courses;
- The student has passed the internship (CU75035)

Brief description of course content:

In this course the student will do a complex project in a small group under coaching of lecturers and experts. The project and professional products will be specific for the study track. The form and account of the results are similar with the graduation phase.

Course learning outcomes:

Test 1:

2.1H: you clarify the company's current situation through coordinated KPIs and an obtained data set and you make an inventory of where the company can still take steps for improvement. Taking into account improvements in, among other things, new technologies

2.2E: you analyse the IST of the processes within the company and you come up with realistic improvement proposals based on the various models and your own vision (SOLL)

2.3C: you realise and evaluate an implementation (plan) based on your own design, so the company has a ready made plan to follow through with the implementation of the changes

2.3D: you describe (and carry out if possible) a relevant change management method and strategy in which you help the employees with the changes they are about to encounter so that you can help resolve possible resistance

2.4C: you advise in a well-argued manner the best option for change based on your own vision/core values, a theoretical change model and the core values of the company.

2.5B: you manage the company processes and ensure that they grow with the company or that there is a plan with which these processes are kept up-to-date, taking into account updating, design, maintenance and quality assurance

Test 2:

7.1M: you can employ the right professional skills to complete a project successfully in a complex environment

7.1N: you account for the choices made regarding the professional skills employed

7.3M: As a project group you deliver structured products and account for everyone's role within the project, the method followed and evaluate the process and the product critically

Test 3:

7.3L: As a project group you can report and present professionally, both verbally and in a report

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Groupassignment	60%	5.5	S2.18 & S2.20
TOETS02 (VT)	Portfolio	Portfolio (individual)	20%	5.5	S2.16 & S2.19
TOETS03 (VT)	Criterion-referenced interview	Criterium focused interview	20%	5.5	S2.19 & S2.20

Block / Semester: J	
CU75051V1	**Title: Graduation Business IT Consultant**

Course information

Amount of study credits:	**Language:**
30	English Dutch

Conditions for course participation:

- The student is in possession of the propaedeutic certificate of the HBO-ICT program;
- The student has obtained at least 137.5 EC from the main phase with completed courses.

Conditions for test participation:

As included in Graduation Student Manual on the graduation Learn page.

Brief description of course content:

Students conduct their graduation on a complex practical assignment in a complex situation. The students does this independently. The final products are qualitative sufficient professional software engineering products, supplemented with an account of the methodical and professional approach. Final results will be presented followed by an assessment of two examiners and possibly one external expert.

Course learning outcomes:

Test 1:

you can independently in a complex environment employ the right professional skills to complete a project successfully
7.1P: you account for the choices made regarding the professional skills employed
7.3O: You deliver structured products, account for the method followed and evaluate the process and the product critically
You can independently make a validated process analysis (IST) for the ICT provisions in a complex context
2.2F: You can independently make a validated and considered process design (SOLL) in a complex context
2.3E: You independently realise an implementation(plan) and test the acceptance in a complex context
2.4D: You can independently give a sound organizational advice for implementing ICT possibilities in a complex context
2.5.C: You can independently draw up a control plan for ICT processes in a complex context.

Test 2:

7.3N: You can report and present professionally, both verbally and in a report

Assessment information

Test code	**Assessment type**	**Assessment description**	**Weighting Factor** (%)	**Minimum score**	**Test opportunities** (block codes)
TOETS01 (VT)	Portfolio	Portfolio+assessment	80%	5.5	S1.7, S1.10, S1.17, S1.20, S2.7, S2.10, S2.17, S2.20
TOETS02 (VT)	Criterion-referenced interview	Criterium focused interview	20%	5.5	S1.9, S1.10, S1.19, S1.20, S2.9, S2.10, S2.19, S2.20

Program profiles for the tracks from cohort 2017-2018 and newer.

Program profile for SE track

	Analysis	Design	Realisation	Advise	Manage & Control
User Interaction	2	2	2		
Organisational Processes	2	1		2	
Infrastructure		2	1	2	2
Software	3	3	3	3	3
Hardware Interfacing	1				
Data Science	2	2	2	2	-
Professional Skills	3	2	3	3	

Program profile for DS track

	Analysis	Design	Realisation	Advise	Manage & Control
User Interaction	2	2	2	2	
Organisational Processes	2	1		2	
Infrastructure		2	1	2	2
Software	2	2	1-2		3
Hardware Interfacing	1				
Data Science	3	3	3	3	-
Professional Skills	3	2	3	3	

Program profile for BIC track

	Analysis	Design	Realisation	Advise	Manage & Control
User Interaction	2	2	2	2	
Organisational Processes	3	3	2	3	3
Infrastructure			1		2
Software	2	2	1		3
Hardware Interfacing	1				
Data Science	2	2	2	2	-
Professional Skills	3	3	3	3	